Michael Galileo Hasian (201344020)	**Rangkuman Minggu ke-5**	Senin, 26 September 2022
		Aplikasi Mikrokontroler
3-TNK		Ferry Satria, BSEE, M.T. Rahmawati Hasanah, S.ST.,M.T.

Aplikasi Interrupts

Perintah untuk meng-*enable external interrupts* secara umum →SEI → *Set Interrupts*
Perintah untuk meng-*disable external interrupts* secara umum →CLI → *Clear Interrupts*

Jika kita menggunakan suatu *interrupts* dalam program kita, maka untuk memastikan bahwa *interrupts* tidak akan menganggu bagian dari program lainnya kecuali saat *interrupts* tersebut diaktifkan.

Contoh Program Aplikasi *Interrupts*
Buat program untuk menampilkan huruf "A" pada unit 7-segment common anode. Jika tombol INT 0 ditekan akan menampilkan "a" selama 2 detik

- **Kode Program**

```
.INCLUDE"M8535DEF.INC"
.ORG 0
RJMP MAIN

.ORG 1
RJMP INT_0

MAIN:   LDI R16,LOW(RAMEND)
        OUT SPL,R16
        LDI R16,HIGH(RAMEND)
        OUT SPH,R16

        CLI
```

```
        LDI R19,0X40
        OUT GICR,R19          ; GICR HANYA MENGENABLE INT0
        LDI R20,3
        OUT MCUCR,R20         ; INT0 AKTIF SAAT RISING EDGE
        SEI

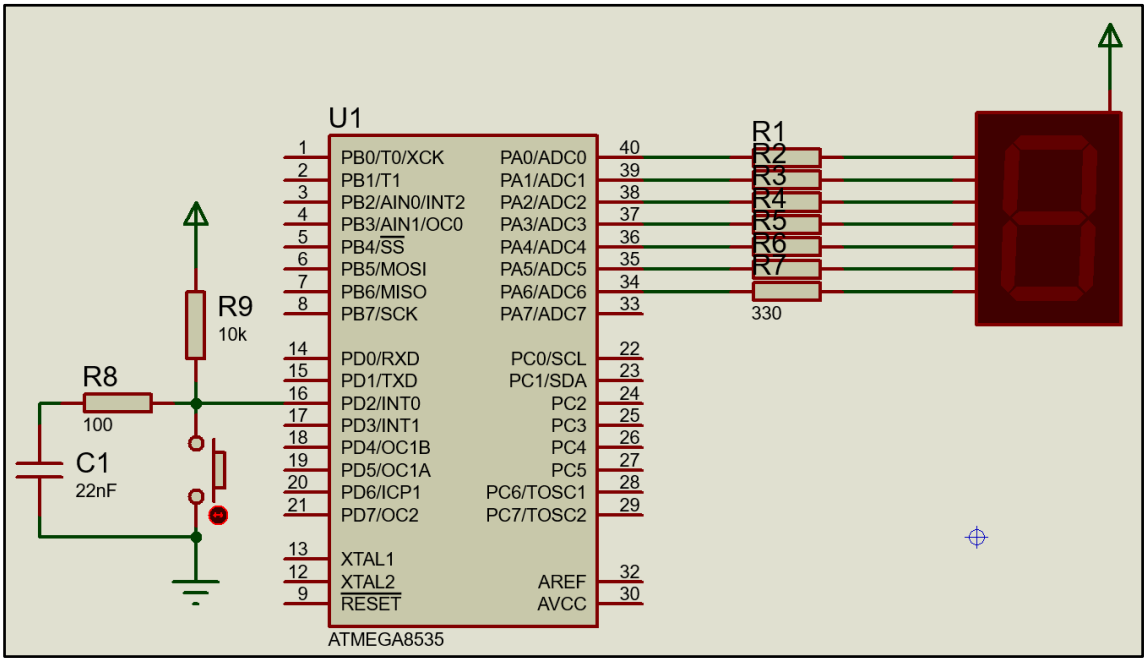
        LDI R17,0XFF
        OUT DDRA,R17          ; PORT B SEBAGAI OUTPUT

        LDI R18,8             ; A KAPITAL PADA 7 SEGMENT CA
        OUT PORTA,R18

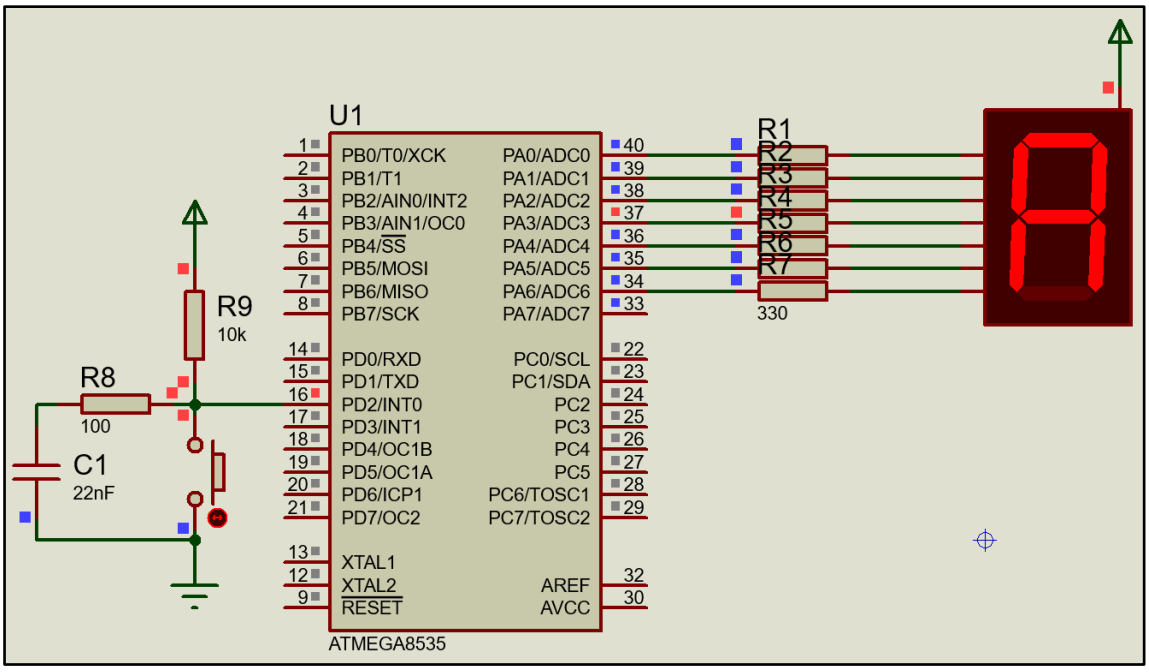
DELAY:  CLI
        LDI R23,3
LUP1:   LDI R24,0XFF
LUP2:   LDI R25,0
LUP3:   DEC R25
        BRNE LUP3
        DEC R24
        BRNE LUP2
        DEC R23
        BRNE LUP1
        SEI
        RET

INT_0:  CLI
        LDI R26,0X20          ; A KECIL PADA 7 SEGMENT CA
        OUT PORTA,R26
        RCALL DELAY
        RCALL DELAY
        SEI
        RETI
```

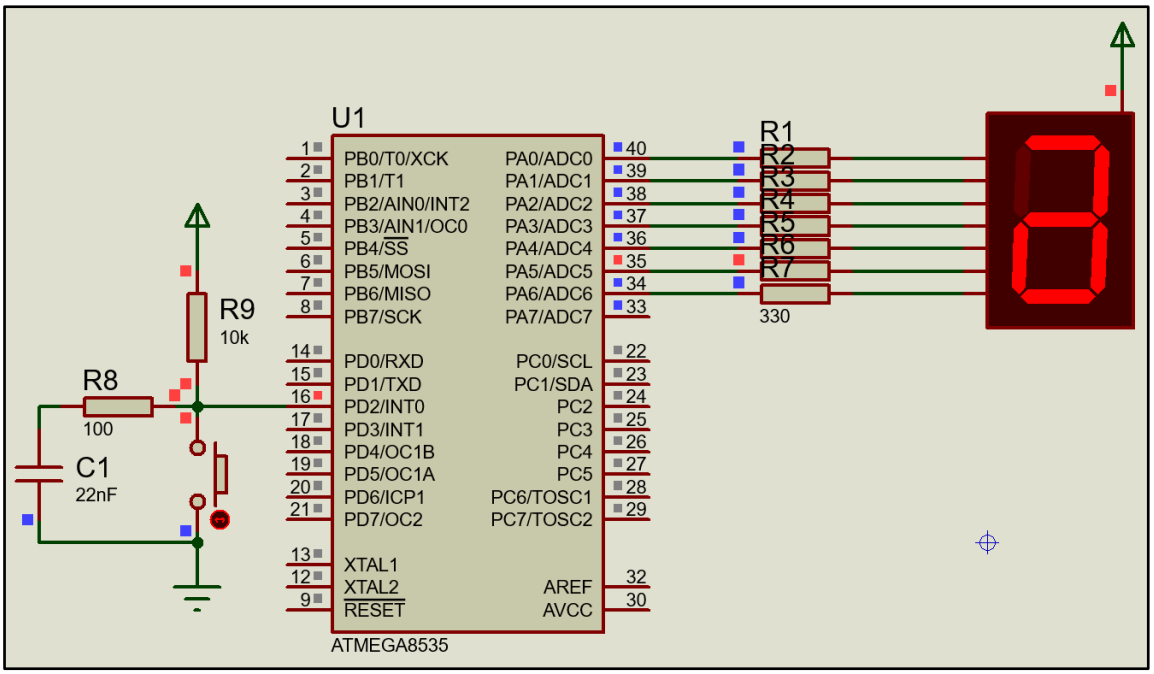
- **Rangkaian Simulasi**



- **Output Program**



Kondisi saat *interrupts* tidak diaktifkan



Kondisi saat *interrupts* diaktifkan

Buat program untuk menampilkan kata belajar pada unit 7-segment common anode. Jika tombol INT 0 ditekan akan menyebabkan kata belajar kedap-kedip.

- **Kode Program**

```
;PROGRAM INTERRUPT DAN TIMER
; PAKAI PROJECT "ROOT"
.INCLUDE "M8535DEF.INC"
.ORG 0X00
.EQU TIMER_1=0XFFF0
RJMP MAIN

.ORG 0X001
RJMP INT_1

MAIN :
    LDI R16,LOW(RAMEND)
    OUT SPL,R16
    LDI R16,HIGH(RAMEND)
    OUT SPH,R16
```

```
HASNA:
    CLI
    LDI R17,0X00
    OUT GICR,R17  ; DISABLE SEMUA INTERRUPT

    LDI R17,0X0C
    OUT MCUCR,R17  ;PILIH SISI NAIK ATAU SISI TURUN

    LDI R17,0X40  ; ENABLE GENERAL INT
    OUT GICR,R17

    LDI R17,0XFF   ; ATUR PORTB DAN PORTC SBG OUTPUT
    OUT DDRB,R17
    OUT DDRC,R17

    SEI    ; SET GENERAL INT

    CLR R20  ; R20 COUNTER

;=====================
LOOP:
    LDI ZL,LOW(2*POSISI)
    LDI ZH,HIGH(2*POSISI)

    LDI R26,$60  ; INISIALISASI POINTER X
    CLR R27

LUP:
    LPM R18,Z+  ; DATA POSISI DIPINDAHKAN KE DATA AREA
    ST X+,R18
    CPI R26,$68
    BRNE LUP

;======================
```

```
LUP2:
    RCALL NYALA0
    ANDI R20,0X01 ; JIKA R20 GENAP, TDK BERKEDIP
    BREQ LUP2    ; JIKA R20 GANJIL, BERKEDIP
    RCALL MATI0
    RJMP LUP2

NYALA0:
    CLI   ; CLEAR INT
    LDI R29,$0F   ; R29=15

NYALA:
    LDI R26,$60   ;POINTER X MENUNJUK KE DATA AREA
    CLR R27      ; ------------------------------
    LDI ZL,LOW(2*POLA) ; POINTER Z MENUNJUK KE POLA KARAKTER
    LDI ZH,HIGH(2*POLA)

NYALA1:
    LD R28,X+
;   COM R28
    OUT PORTB,R28

    LPM R21,Z+
    OUT PORTC,R21

    RCALL DELAY_2MS
    CPI R26,$68
    BRNE NYALA1

    DEC R29
    BRNE NYALA
    SEI
    RET
;==========================
```

```
;============================
MATI0:
    CLI
    LDI R29,125

MATI:
    LDI R21,0XFF
    OUT PORTC,R21

    RCALL DELAY_2MS

    DEC R29
    BRNE MATI
    SEI
    RET
;============================

DELAY_2MS:
    CLI
    PUSH R16
    PUSH R17
    LDI R16,4
    OUT TIMSK,R16
    LDI R16, HIGH(TIMER_1)
    OUT TCNT1H,R16
    LDI R16, LOW(TIMER_1)
    OUT TCNT1L,R16
    LDI R16,5
    OUT TCCR1B,R16

LOOPTIMER:
  IN R17,TIFR
  SBRS R17,TOV1
```

RJMP LOOPTIMER

LDI R16,4

OUT TIFR,R16

POP R17

POP R16

SEI

RET

INT_1:

CLI

INC R20

SEI

RETI

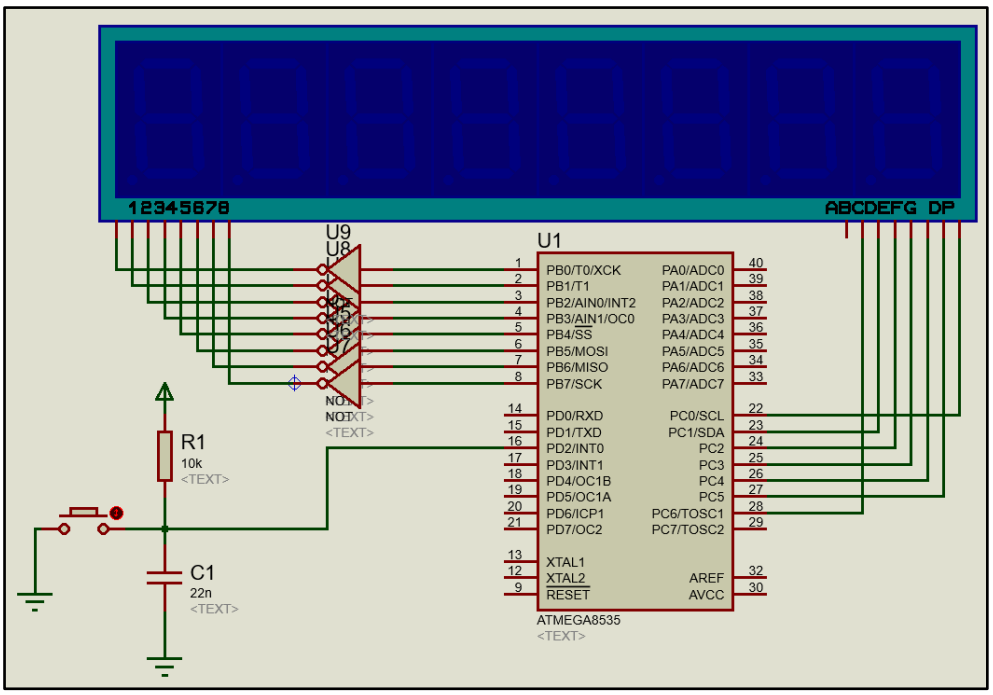
POLA:

.DB 0X83, 0X84, 0XC7, 0XA0, 0XF0, 0XA0, 0XA0, 0XBF

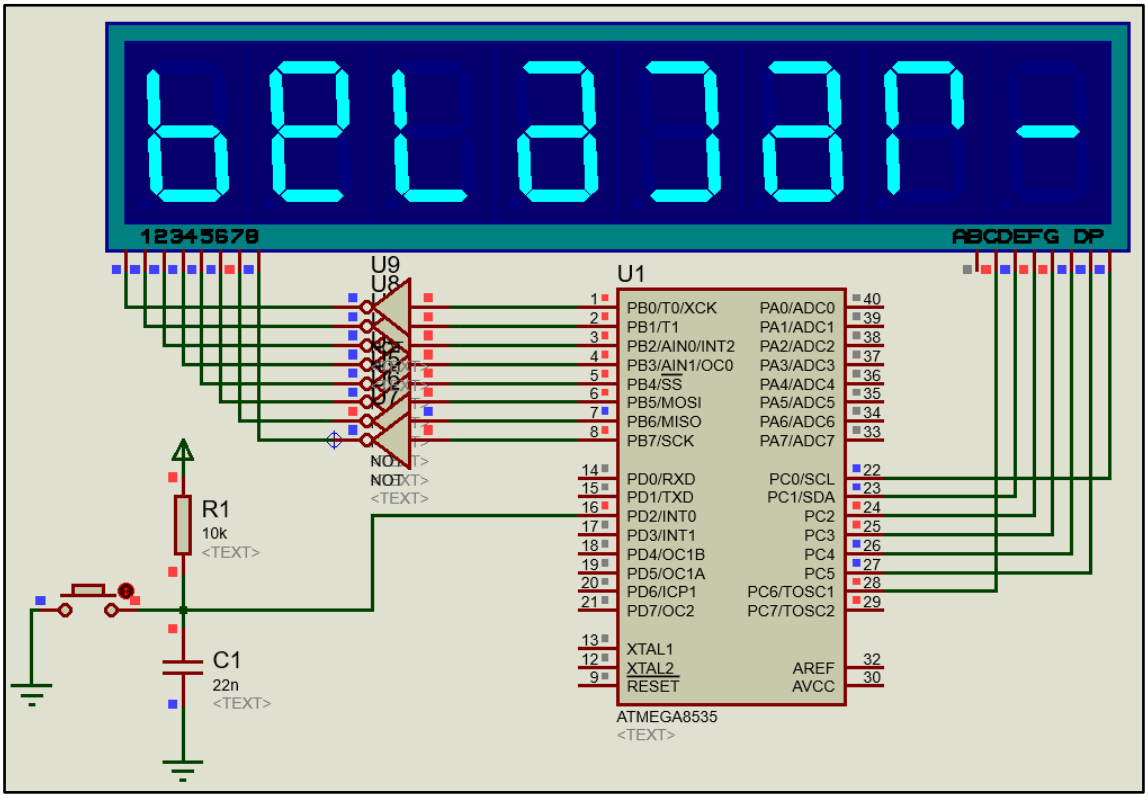
POSISI:

.DB 0XFE, 0XFD, 0XFB, 0XF7, 0XEF, 0XDF, 0XBF, 0X7F

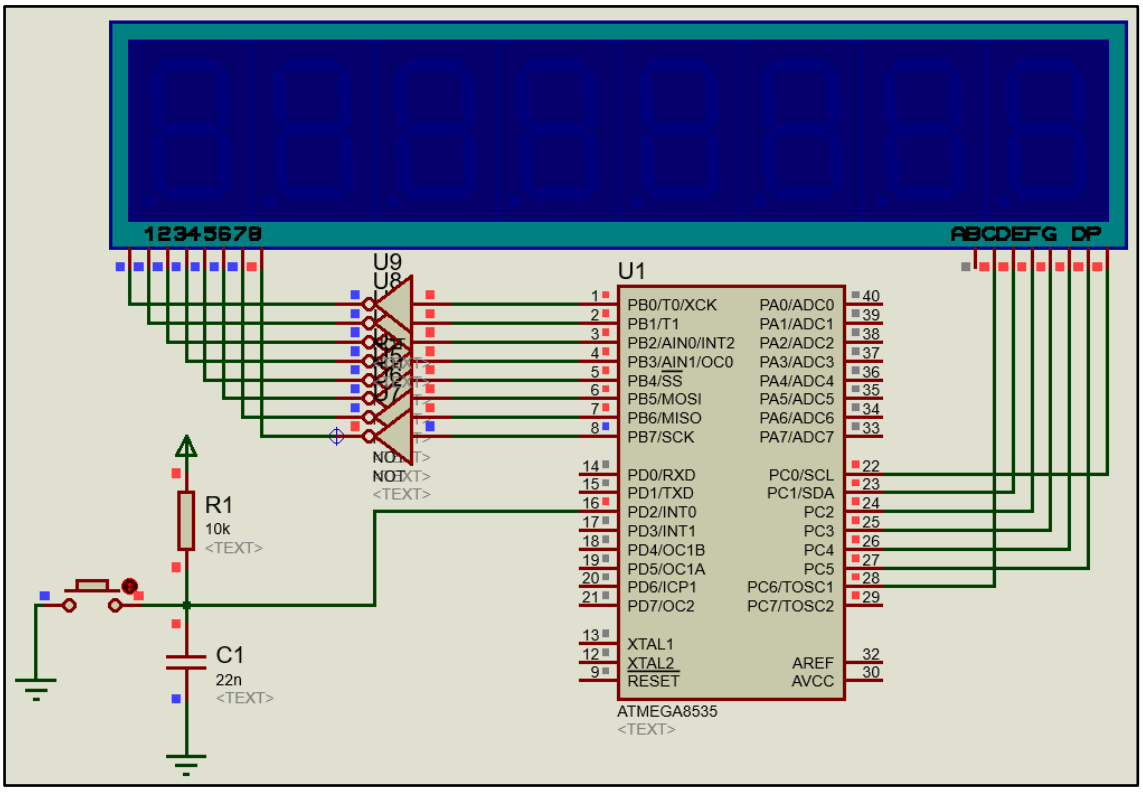
- **Rangkaian Simulasi**

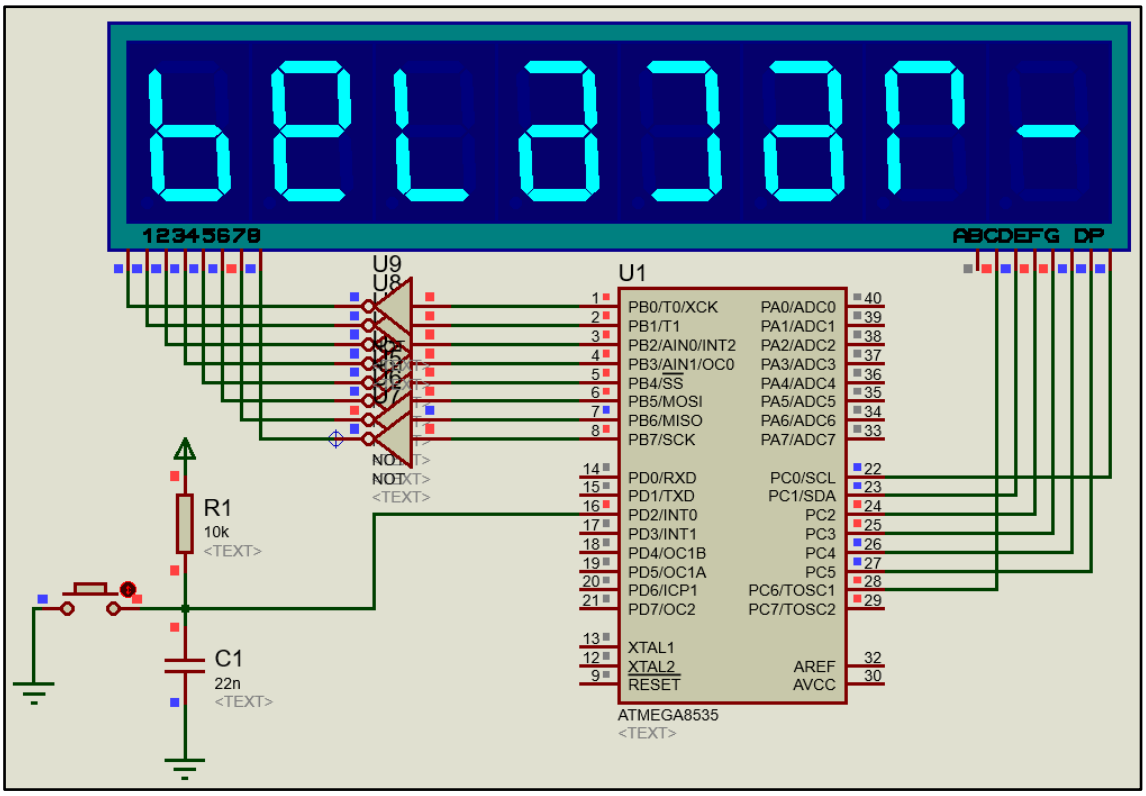


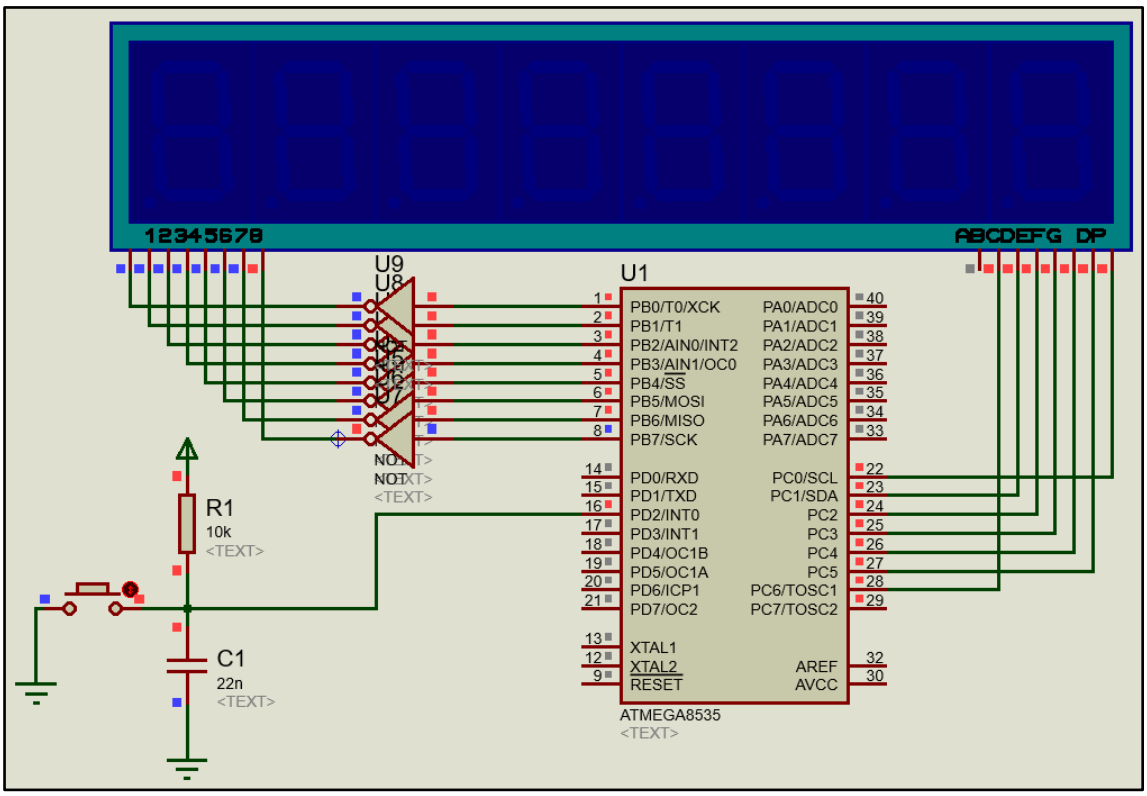
- **Output Program**

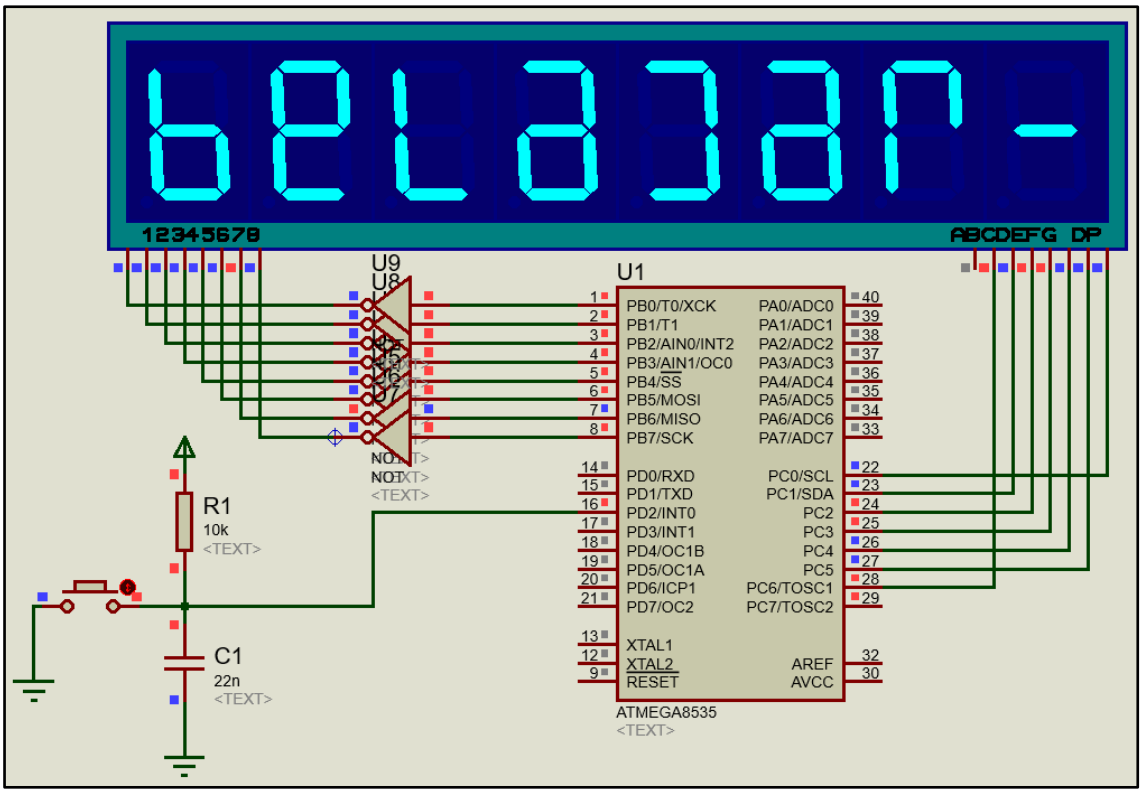


Kondisi saat *interrupts* tidak diaktifkan




12345678
ABCDEFG DP
U9
U8
U1
PB0/T0/XCK
PB1/T1
PB2/AIN0/INT2
PB3/AIN1/OC0
PB4/SS
PB5/MOSI
PB6/MISO
PB7/SCK
PD0/RXD
PD1/TXD
PD2/INT0
PD3/INT1
PD4/OC1B
PD5/OC1A
PD6/ICP1
PD7/OC2
XTAL1
XTAL2
RESET
PA0/ADC0
PA1/ADC1
PA2/ADC2
PA3/ADC3
PA4/ADC4
PA5/ADC5
PA6/ADC6
PA7/ADC7
PC0/SCL
PC1/SDA
PC2
PC3
PC4
PC5
PC6/TOSC1
PC7/TOSC2
AREF
AVCC
ATMEGA8535
R1
10k
C1
22n


12345678
ABCDEFG DP
U9
U8
U1
PB0/T0/XCK
PB1/T1
PB2/AIN0/INT2
PB3/AIN1/OC0
PB4/SS
PB5/MOSI
PB6/MISO
PB7/SCK
PD0/RXD
PD1/TXD
PD2/INT0
PD3/INT1
PD4/OC1B
PD5/OC1A
PD6/ICP1
PD7/OC2
XTAL1
XTAL2
RESET
PA0/ADC0
PA1/ADC1
PA2/ADC2
PA3/ADC3
PA4/ADC4
PA5/ADC5
PA6/ADC6
PA7/ADC7
PC0/SCL
PC1/SDA
PC2
PC3
PC4
PC5
PC6/TOSC1
PC7/TOSC2
AREF
AVCC
ATMEGA8535
R1
10k
C1
22n



Kondisi saat *interrupts* diaktifkan

Soal Lab Minggu Ke-5

1. Rancang suatu Up Counter Heksadesimal 1 digit Modulo 16 (Menghitung naik secara berulang dari 0 sampai F). Gunakan seven segment jenis Common Cathode dengan tombol interupsi INT0 yang berfungsi untuk menaikkan nilai counter (Counter incrementer).

- **Kode Program**

```
.include "m8535def.inc"
.org 0
RJMP MAIN

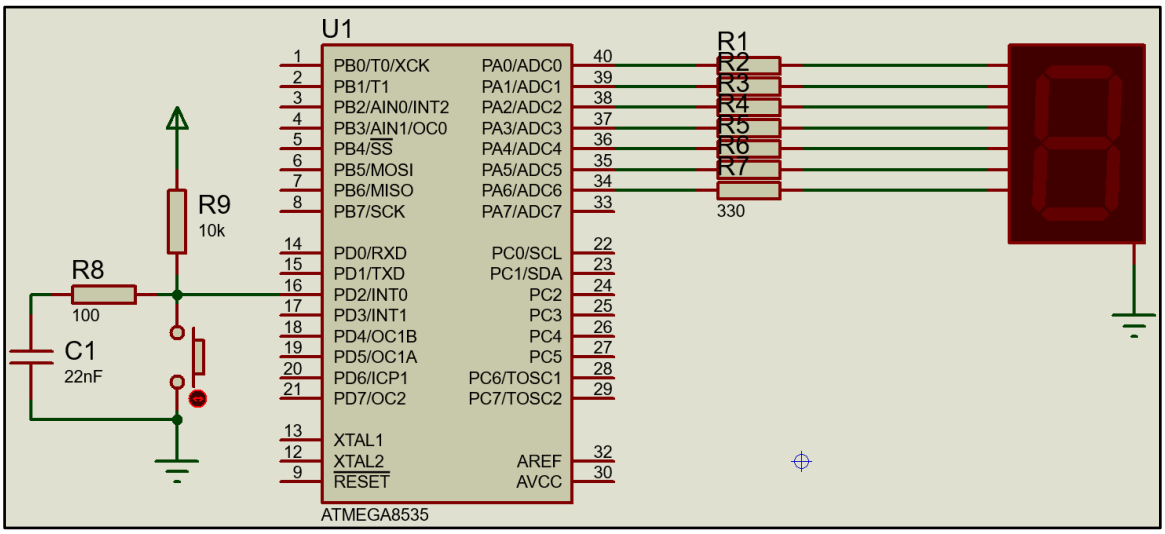
.ORG 0X01
RJMP INT_0
; Up-Counter Heksadesimal Modulo 16 (Menghitung secara berulang
; dari 0 sampai F); INT0 berfungsi sebagai Counter Incrementer
; Menggunakan Seven Segment Common Cathode.
; PROTEUS PROJECT: UP-CNT-INT0
;
```

```
MAIN:   CLI
        LDI R20,LOW(RAMEND)
        OUT SPL,R20
        LDI R20,HIGH(RAMEND)
        OUT SPH,R20
        LDI R20,0XFF
        OUT DDRA,R20
        LDI R20,0X40
        OUT GICR,R20
        LDI R20,0X02
        OUT MCUCR,R20
        LDI R30,LOW(2*POLA) ; INIT POINTER Z
        LDI R31,HIGH(2*POLA)
        LDI R22,0 ; INIT COUNTER
        SEI
KK:             MOV R30,R22
        LPM R2,Z
        OUT PORTA,R2
        RCALL DELAY
        RJMP KK

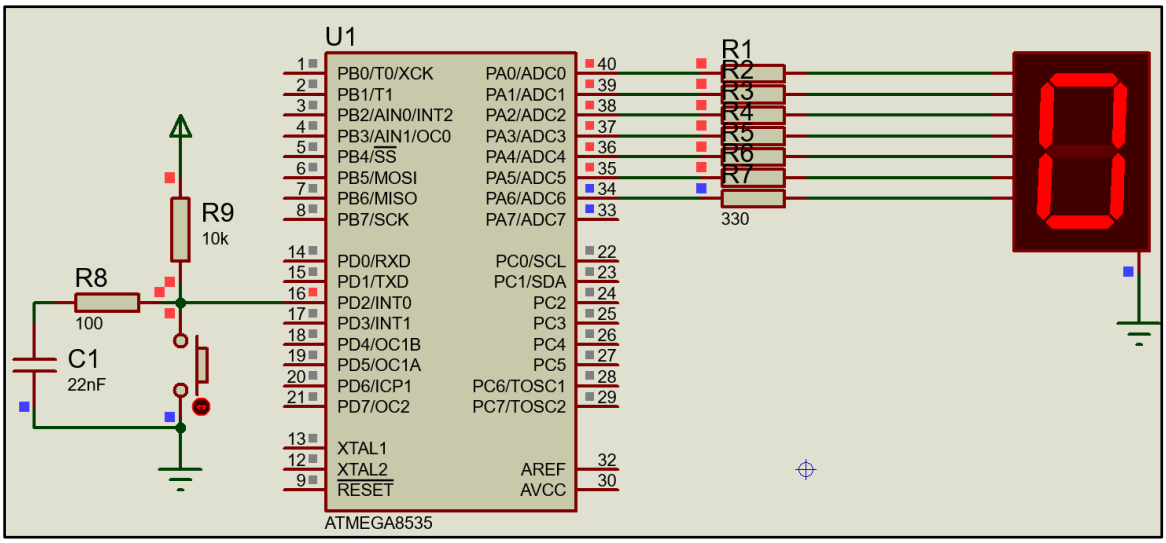
DELAY:  CLI
        LDI R23,1
LP1:    LDI R24,4
LP2:    LDI R25,0
LP3:    DEC R25
        BRNE LP3
        DEC R24
        BRNE LP2
        DEC R23
        BRNE LP1
        SEI
        RET
        ;=================
```

```
INT_0:      ;CLI
            INC R22
            CPI R22,0X10
            BRNE LK1
            LDI R22,0
LK1:;       SEI
            RETI
;==================
.ORG 0X80
POLA:
.DB 0X3F,0X06,0X5B,0X4F,0X66,0X6D,0X7D,0X07
.DB 0X7F,0X6F,0X77,0X7C,0X39,0X5E,0X79,0X71
```

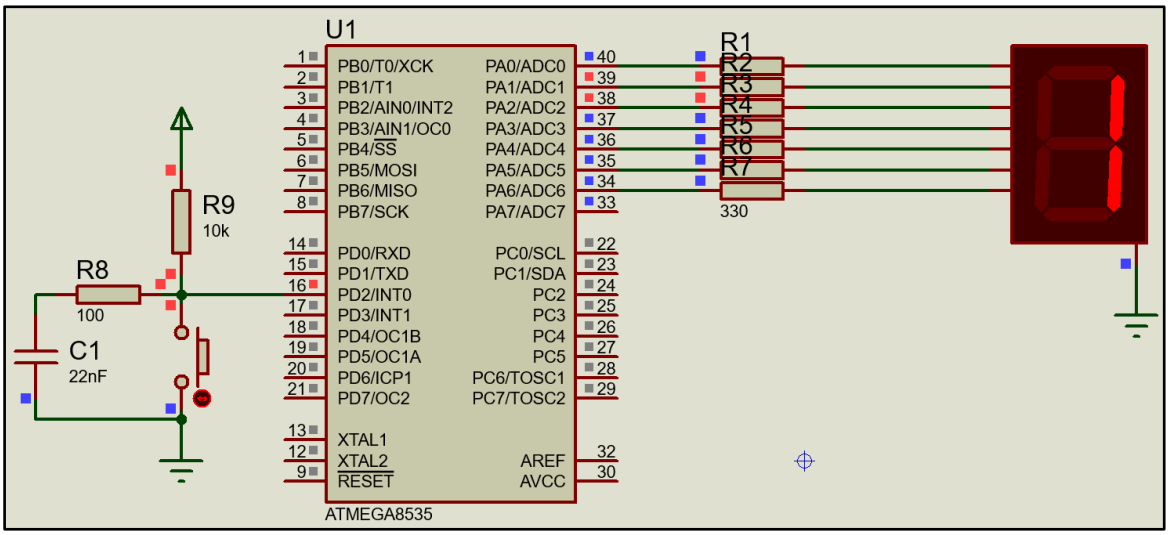
- **Rangkaian Simulasi**



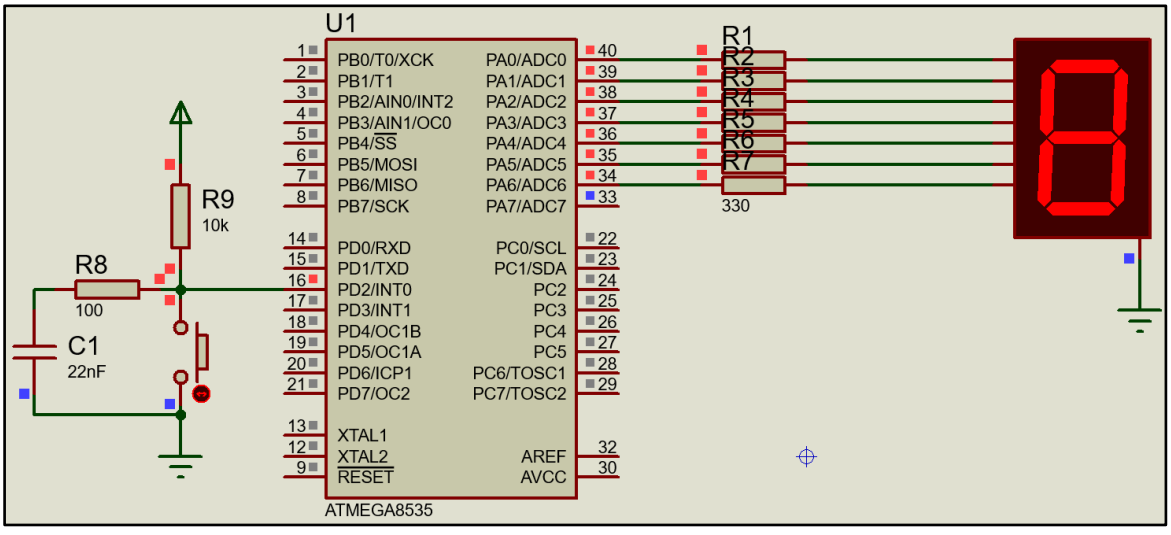
- **Ouput Program**

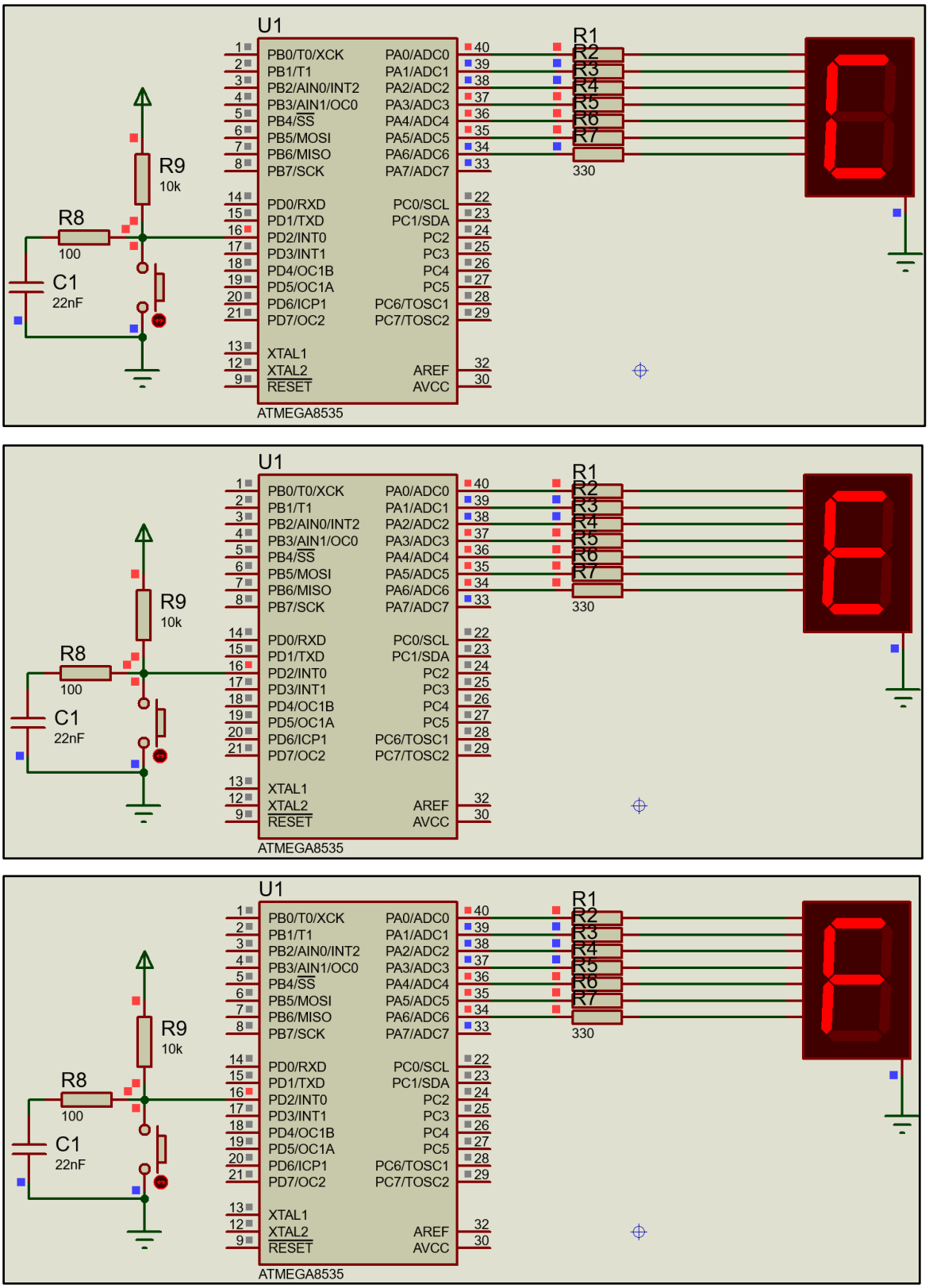


Kondisi awal *counter*



Kondisi ketika *push button* ditekan sekali





Counter akan terus menghitung naik ketika *push button* ditekan

2. Seperti soal nomor 1, tapi untuk Down Counter.

- **Kode Program**

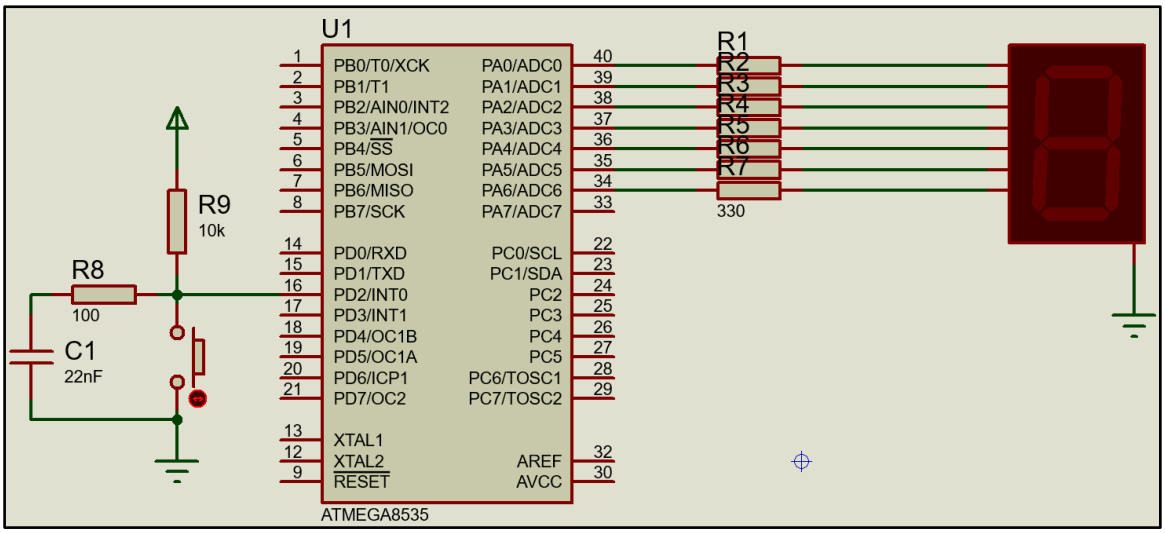
```
.include "m8535def.inc"
.org 0
RJMP MAIN

.ORG 0X01
RJMP INT_0
; Down-Counter Heksadesimal Modulo 16 (Menghitung secara berulang
; dari F sampai 0); INT0 berfungsi sebagai Counter Incrementer
; Menggunakan Seven Segment Common Cathode.
; PROTEUS PROJECT: UP-CNT-INT0
;

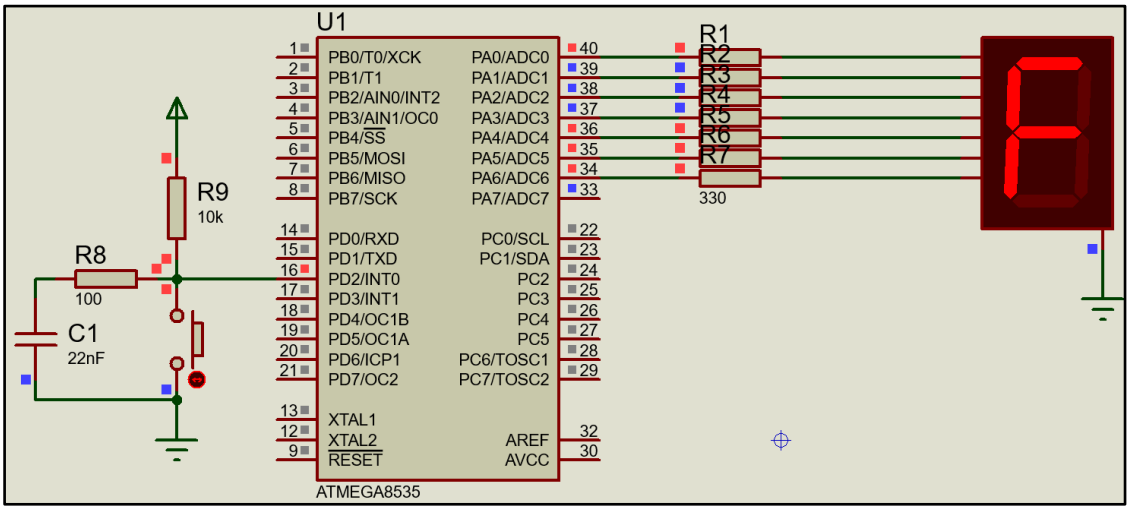
MAIN:   CLI
        LDI R20,LOW(RAMEND)
        OUT SPL,R20
        LDI R20,HIGH(RAMEND)
        OUT SPH,R20
        LDI R20,0XFF
        OUT DDRA,R20
        LDI R20,0X40
        OUT GICR,R20
        LDI R20,0X02
        OUT MCUCR,R20
        LDI R30,LOW(2*POLA) ; INIT POINTER Z
        LDI R31,HIGH(2*POLA)
        LDI R22,0                   ; INIT COUNTER
        SEI
KK:     MOV R30,R22
        LPM R2,Z
        OUT PORTA,R2
        RCALL DELAY
        RJMP KK
```

```
DELAY:  CLI
        LDI R23,1
LP1:    LDI R24,4
LP2:    LDI R25,0
LP3:    DEC R25
        BRNE LP3
        DEC R24
        BRNE LP2
        DEC R23
        BRNE LP1
        SEI
        RET
        ;=================
INT_0:  ;CLI
        INC R22
        CPI R22,0X10
        BRNE LK1
        LDI R22,0
LK1:;   SEI
        RETI
;==================
.ORG 0X80
POLA:
.DB 0X71,0X79,0X5E,0X39,0X7C,0X77,0X6F,0X7F
.DB 0X07,0X7D,0X6D,0X66,0X4F,0X5B,0X06,0X3F
```

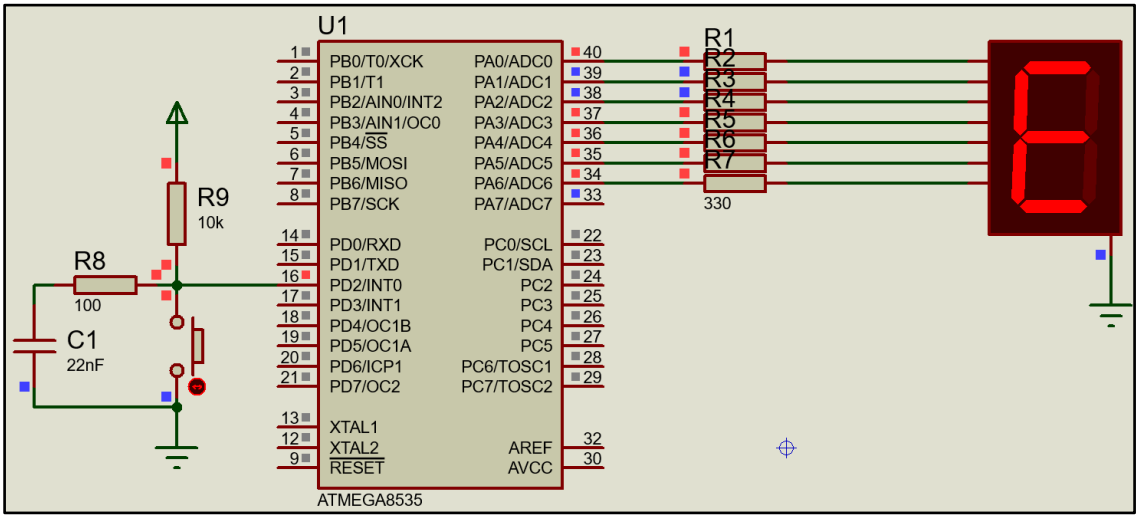
- **Rangkaian Simulasi**



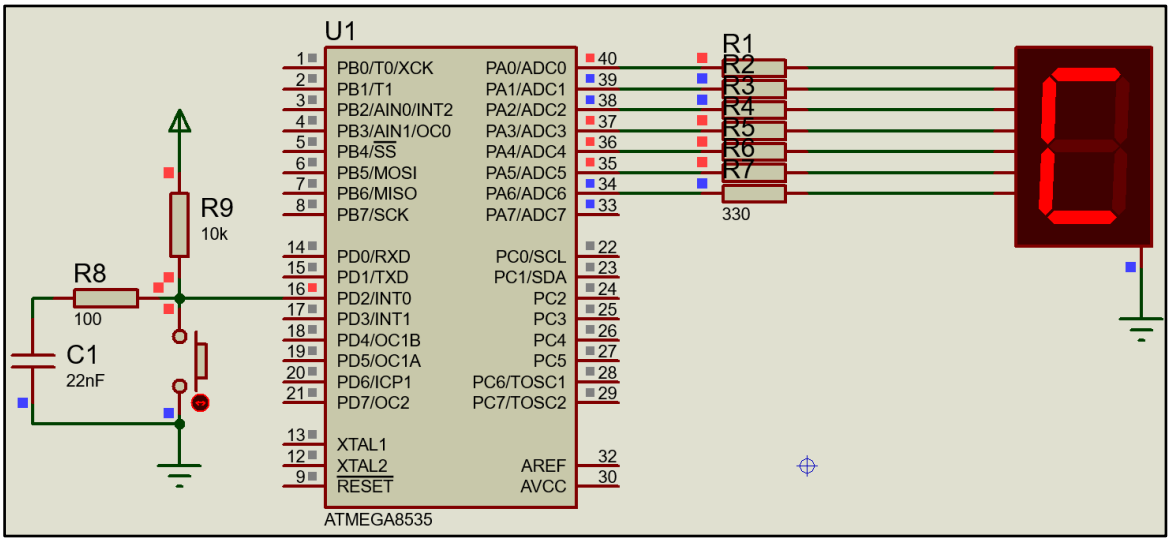
- **Ouput Program**

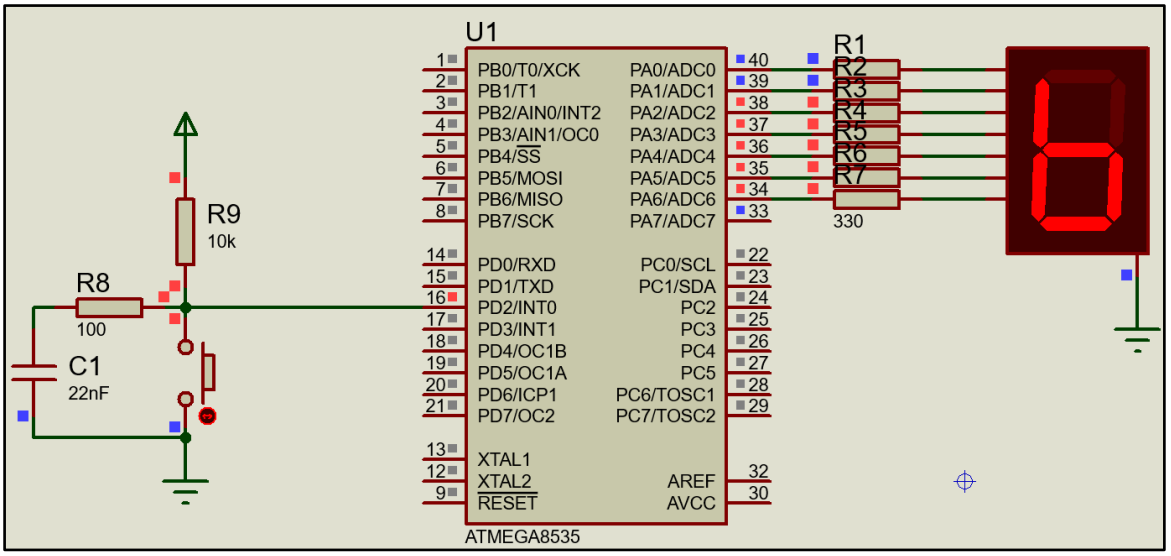


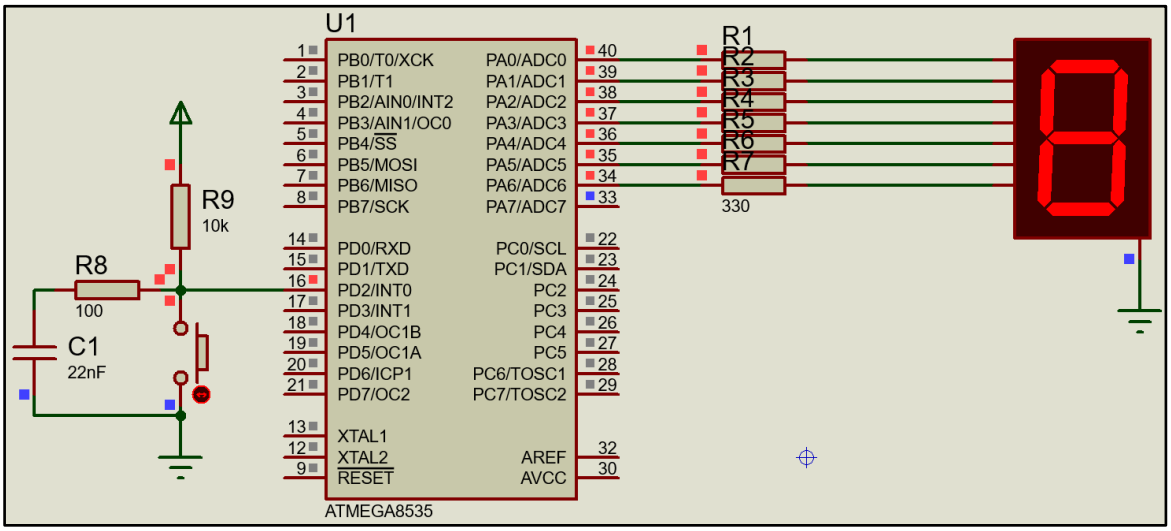
Kondisi awal *counter*

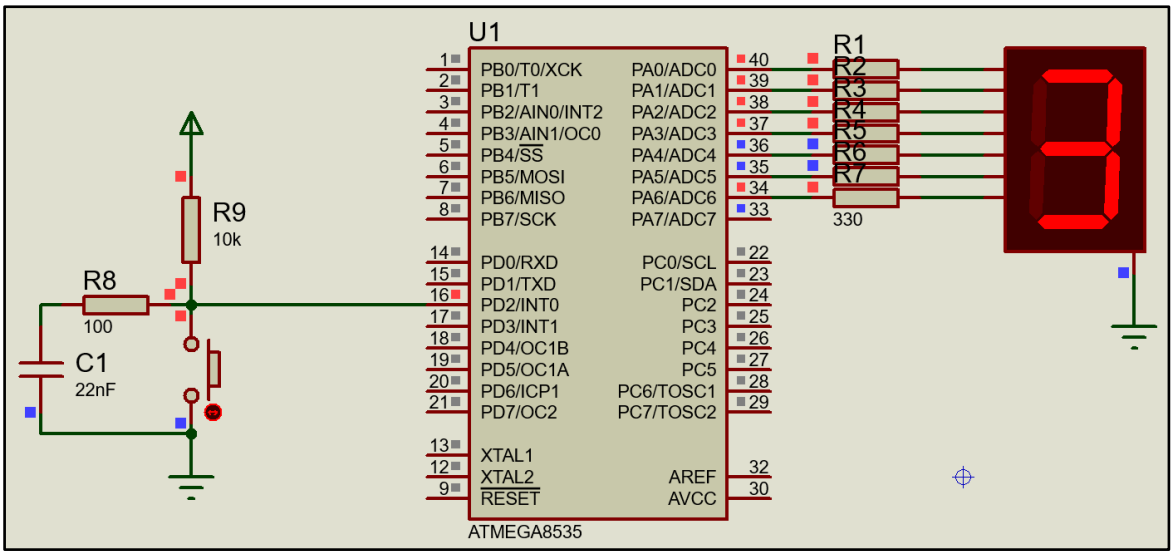


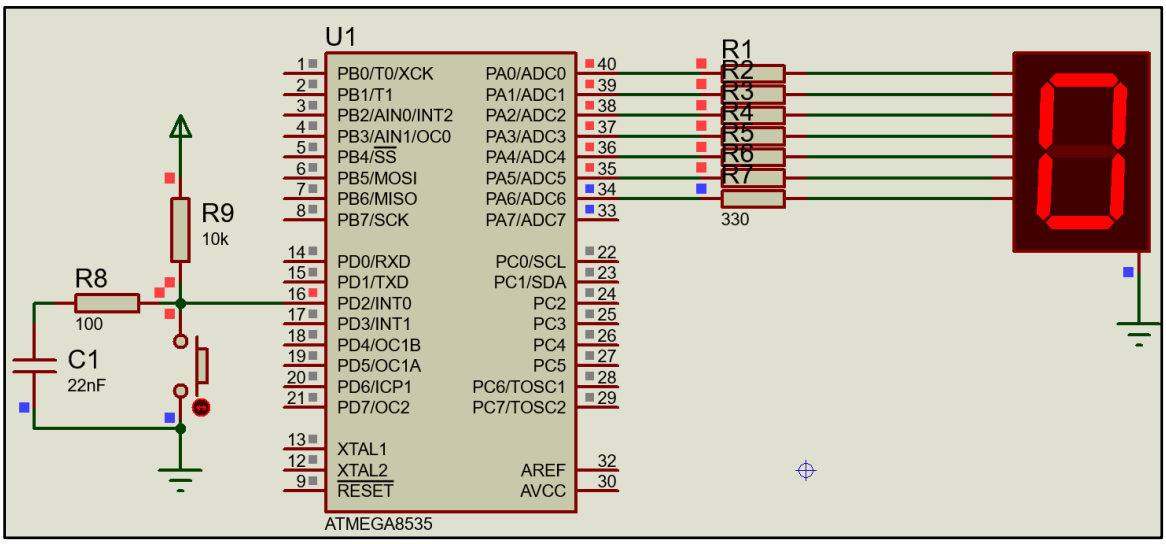
Kondisi ketika *push button* ditekan sekali


U1
PB0/T0/XCK
PB1/T1
PB2/AIN0/INT2
PB3/AIN1/OC0
PB4/SS
PB5/MOSI
PB6/MISO
PB7/SCK
PD0/RXD
PD1/TXD
PD2/INT0
PD3/INT1
PD4/OC1B
PD5/OC1A
PD6/ICP1
PD7/OC2
XTAL1
XTAL2
RESET
PA0/ADC0
PA1/ADC1
PA2/ADC2
PA3/ADC3
PA4/ADC4
PA5/ADC5
PA6/ADC6
PA7/ADC7
PC0/SCL
PC1/SDA
PC2
PC3
PC4
PC5
PC6/TOSC1
PC7/TOSC2
AREF
AVCC
ATMEGA8535
R1
R2
R3
R4
R5
R6
R7
330
R8
100
R9
10k
C1
22nF


U1
PB0/T0/XCK
PB1/T1
PB2/AIN0/INT2
PB3/AIN1/OC0
PB4/SS
PB5/MOSI
PB6/MISO
PB7/SCK
PD0/RXD
PD1/TXD
PD2/INT0
PD3/INT1
PD4/OC1B
PD5/OC1A
PD6/ICP1
PD7/OC2
XTAL1
XTAL2
RESET
PA0/ADC0
PA1/ADC1
PA2/ADC2
PA3/ADC3
PA4/ADC4
PA5/ADC5
PA6/ADC6
PA7/ADC7
PC0/SCL
PC1/SDA
PC2
PC3
PC4
PC5
PC6/TOSC1
PC7/TOSC2
AREF
AVCC
ATMEGA8535
R1
R2
R3
R4
R5
R6
R7
330
R8
100
R9
10k
C1
22nF


U1
PB0/T0/XCK
PB1/T1
PB2/AIN0/INT2
PB3/AIN1/OC0
PB4/SS
PB5/MOSI
PB6/MISO
PB7/SCK
PD0/RXD
PD1/TXD
PD2/INT0
PD3/INT1
PD4/OC1B
PD5/OC1A
PD6/ICP1
PD7/OC2
XTAL1
XTAL2
RESET
PA0/ADC0
PA1/ADC1
PA2/ADC2
PA3/ADC3
PA4/ADC4
PA5/ADC5
PA6/ADC6
PA7/ADC7
PC0/SCL
PC1/SDA
PC2
PC3
PC4
PC5
PC6/TOSC1
PC7/TOSC2
AREF
AVCC
ATMEGA8535
R1
R2
R3
R4
R5
R6
R7
330
R8
100
R9
10k
C1
22nF





Counter akan terus menghitung turun ketika *push button* ditekan

3. Rancang suatu Up/Down Counter Heksadesimal 1 digit Modulo 16 (Menghitung naik/turun secara berulang dengan range output dari 0 sampai F). Gunakan seven segment jenis Common Cathode dengan tombol interupsi INT0 yang berfungsi untuk menaikkan/menurunkan nilai counter (Counter incrementer/decrementer); dan INT1 yang berfungsi sebagai pengatur Mode kerja Counter (up/down).

- **Kode Program**

```
.include "m8535def.inc"
.org 0
RJMP MAIN

.ORG 0X01
```

```
RJMP INT_0

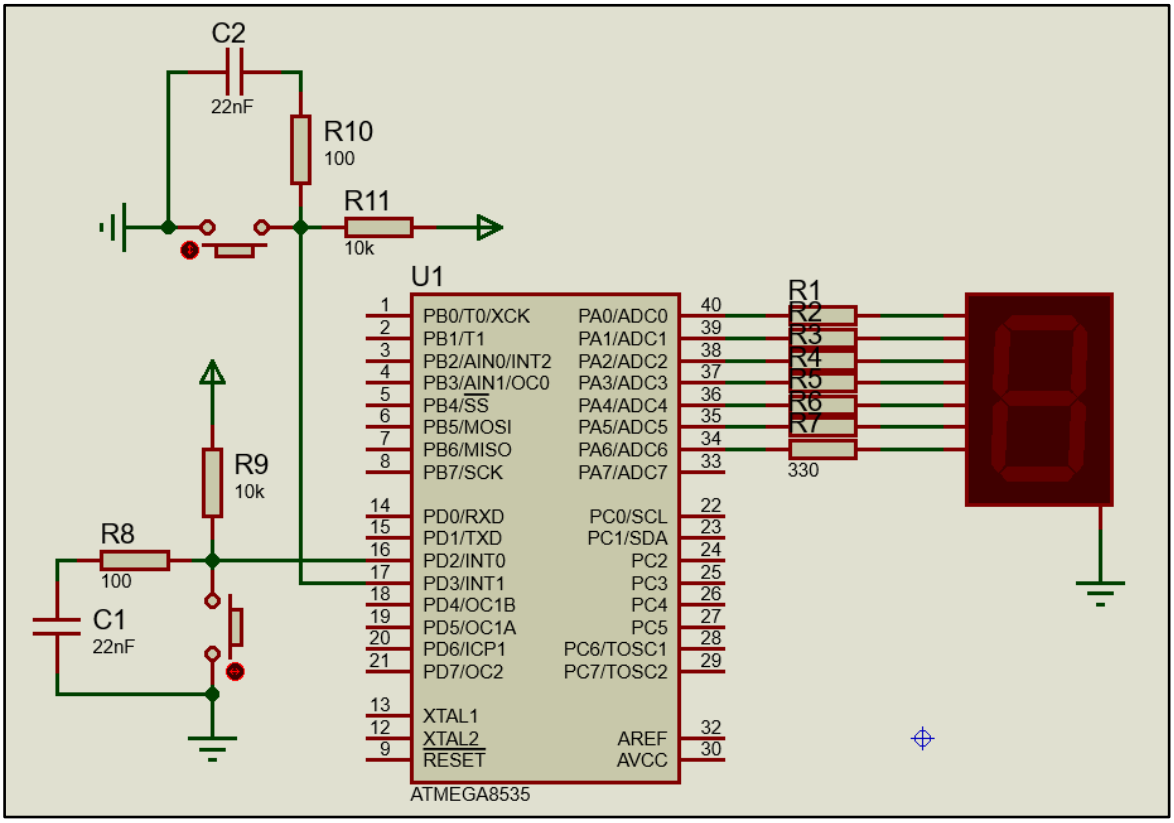
.ORG 0X02
RJMP INT_1
; Up-Down Counter Heksadesimal Modulo 16 (Menghitung secara berulang
; dari 0 sampai F); INT0 berfungsi sebagai Counter Incrementer
; Menggunakan Seven Segment Common Cathode.
; PROTEUS PROJECT: UP-CNT-INT0


MAIN:   CLI
        LDI R20,LOW(RAMEND)
        OUT SPL,R20
        LDI R20,HIGH(RAMEND)
        OUT SPH,R20
        LDI R20,0XFF
        OUT DDRA,R20
        LDI R20,0XC0
        OUT GICR,R20
        LDI R20,0X0F
        OUT MCUCR,R20
        LDI R30,LOW(2*POLA) ; INIT POINTER Z
        LDI R31,HIGH(2*POLA)
        LDI R22,0 ; INIT COUNTER
        SEI
KK:             MOV R30,R22
        LPM R2,Z
        OUT PORTA,R2
        RCALL DELAY
        RJMP KK

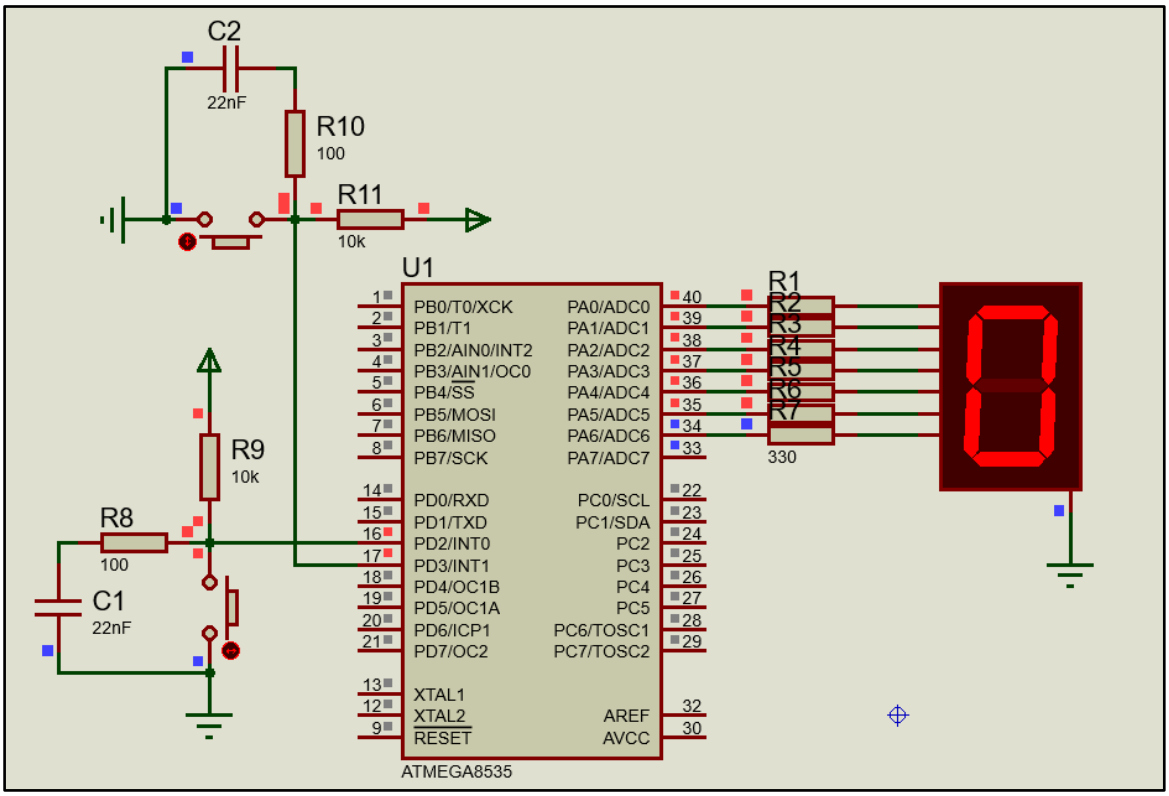
DELAY:  CLI
        LDI R23,1
LP1:    LDI R24,4
```

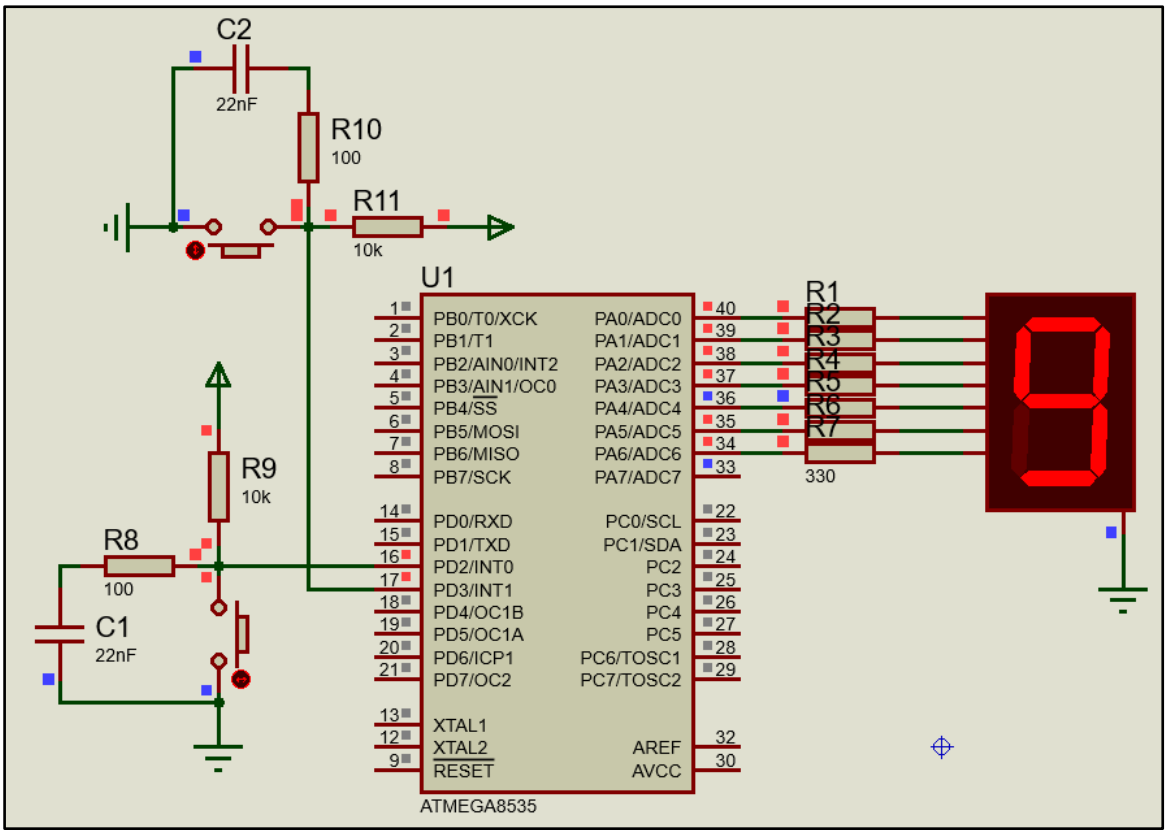
```
LP2:    LDI R25,0
LP3:    DEC R25
        BRNE LP3
        DEC R24
        BRNE LP2
        DEC R23
        BRNE LP1
        SEI
        RET
;=================
INT_0:  CLI
        MOV R21,R20
        ANDI R21,1
        CPI R21,1
        BRNE TRN
        INC R22
        CPI R22,0X10
        BRNE LK1
        LDI R22,0
        RJMP LK1
TRN:    DEC R22
        CPI R22,0XFF
        BRNE LK1
        LDI R22,0X0F
LK1:    SEI
        RETI
INT_1:  CLI
        INC R20
        SEI
        RETI
.ORG 0X80
POLA:
.DB 0X3F,0X06,0X5B,0X4F,0X66,0X6D,0X7D,0X07
.DB 0X7F,0X6F,0X77,0X7C,0X39,0X5E,0X79,0X71
```

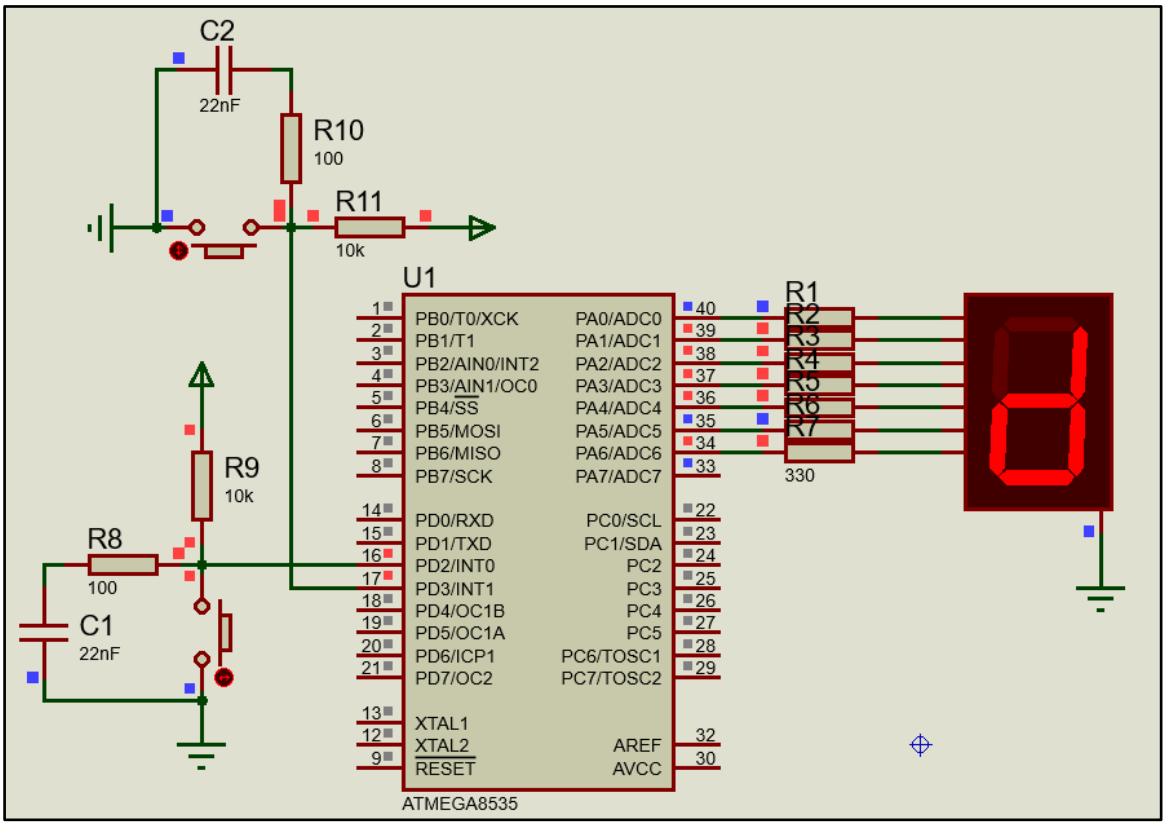
- **Rangkaian Simulasi**

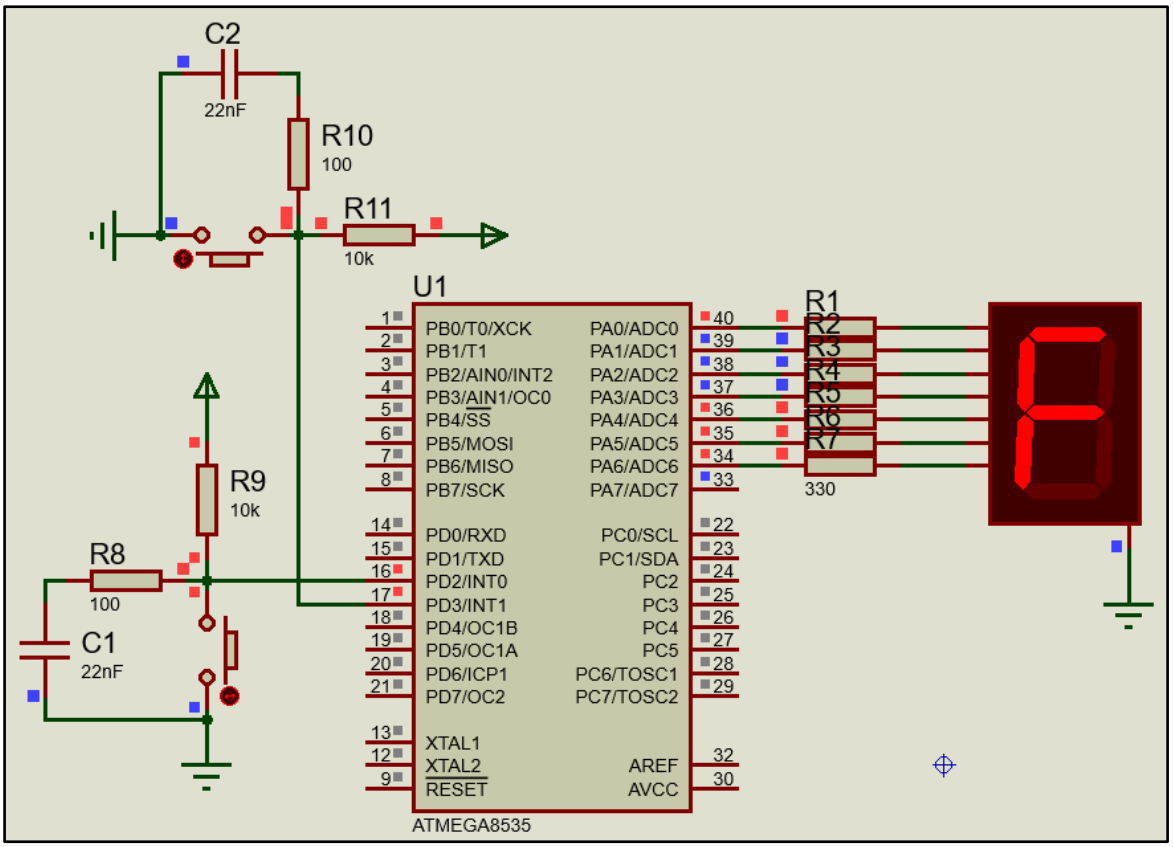


- **Ouput Program**

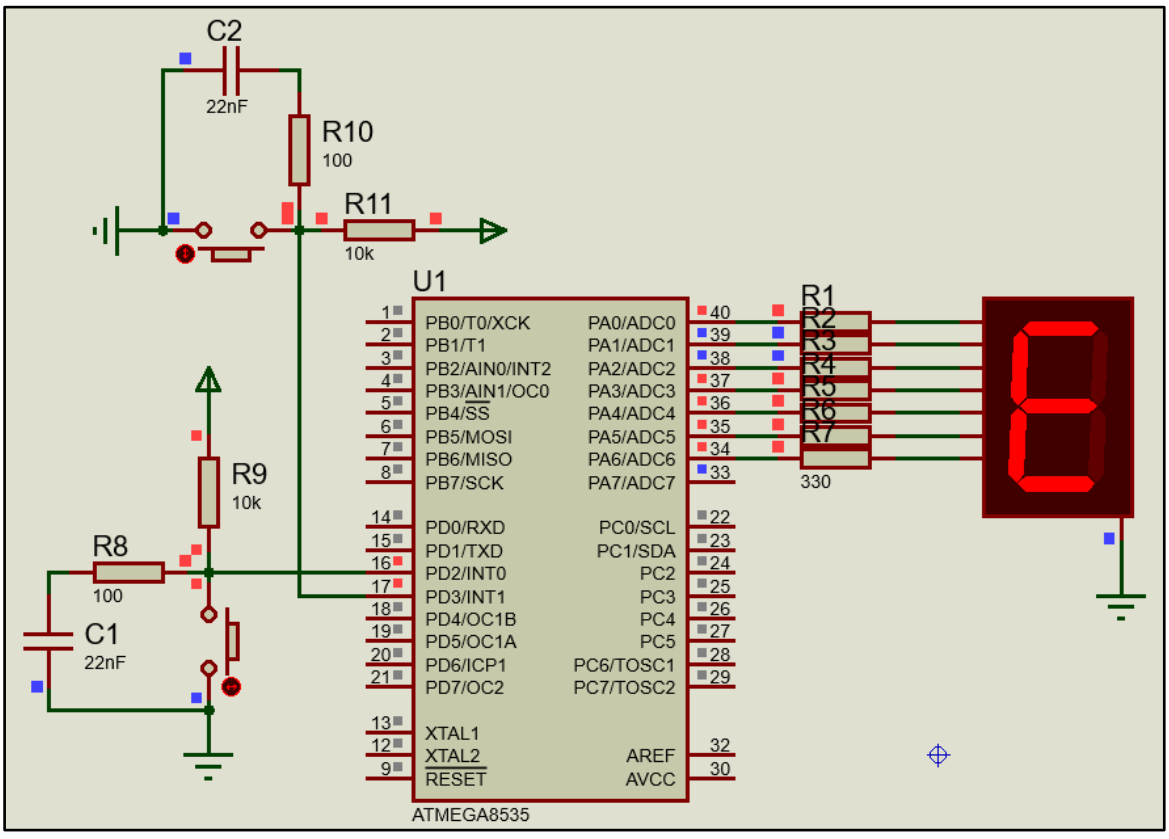


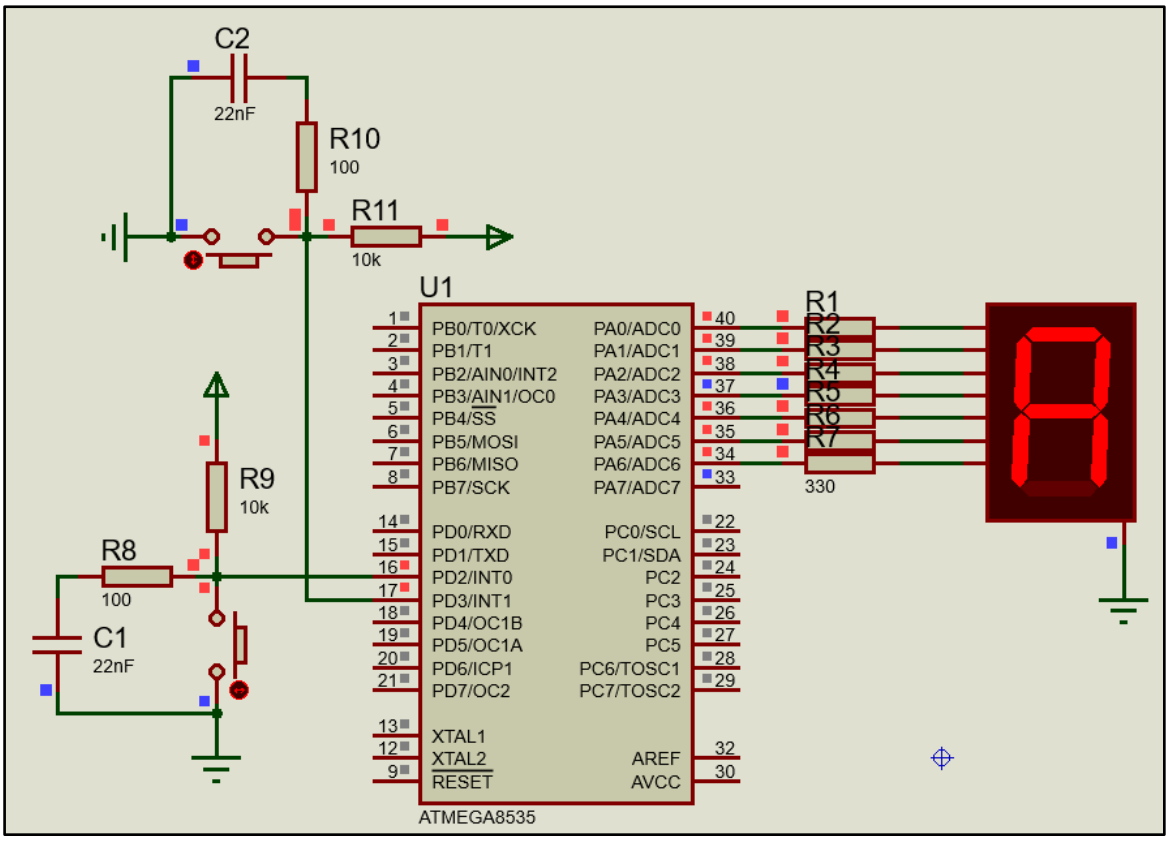

C2
22nF
R10
100
R11
10k
U1
PB0/T0/XCK
PB1/T1
PB2/AIN0/INT2
PB3/AIN1/OC0
PB4/SS
PB5/MOSI
PB6/MISO
PB7/SCK
PD0/RXD
PD1/TXD
PD2/INT0
PD3/INT1
PD4/OC1B
PD5/OC1A
PD6/ICP1
PD7/OC2
XTAL1
XTAL2
RESET
PA0/ADC0
PA1/ADC1
PA2/ADC2
PA3/ADC3
PA4/ADC4
PA5/ADC5
PA6/ADC6
PA7/ADC7
PC0/SCL
PC1/SDA
PC2
PC3
PC4
PC5
PC6/TOSC1
PC7/TOSC2
AREF
AVCC
ATMEGA8535
R1
R2
R3
R4
R5
R6
R7
330
R9
10k
R8
100
C1
22nF

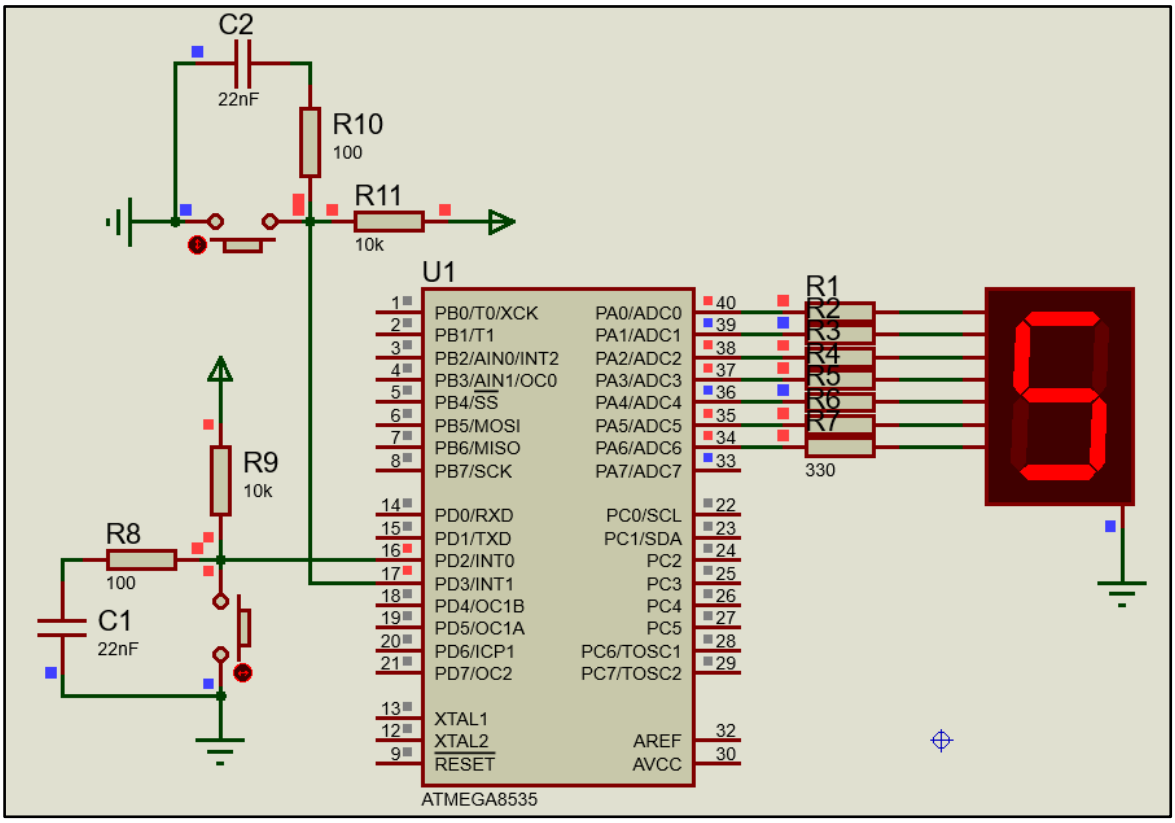

C2
22nF
R10
100
R11
10k
U1
PB0/T0/XCK
PB1/T1
PB2/AIN0/INT2
PB3/AIN1/OC0
PB4/SS
PB5/MOSI
PB6/MISO
PB7/SCK
PD0/RXD
PD1/TXD
PD2/INT0
PD3/INT1
PD4/OC1B
PD5/OC1A
PD6/ICP1
PD7/OC2
XTAL1
XTAL2
RESET
PA0/ADC0
PA1/ADC1
PA2/ADC2
PA3/ADC3
PA4/ADC4
PA5/ADC5
PA6/ADC6
PA7/ADC7
PC0/SCL
PC1/SDA
PC2
PC3
PC4
PC5
PC6/TOSC1
PC7/TOSC2
AREF
AVCC
ATMEGA8535
R1
R2
R3
R4
R5
R6
R7
330
R9
10k
R8
100
C1
22nF



Mode menghitung naik







Mode menghitung turun

4. Rancang suatu Up Counter Heksadesimal 2 digit Modulo 45 (Menghitung naik secara berulang dari 00 sampai 2C). Gunakan seven segment jenis Common Anode dengan tombol interupsi INT2 yang berfungsi untuk menaikkan nilai counter (Counter incrementer).

- **Kode Program**

```
.include "m8535def.inc"
.org 0
RJMP MAIN

.ORG 0X012
RJMP INT_2
; Up-Counter Heksadesimal Modulo 45 (Menghitung secara berulang
; dari 0 sampai 2C); INT0 berfungsi sebagai Counter Incrementer
; Menggunakan Seven Segment Common Anode.
; PROTEUS PROJECT: UP-CNT-INT0
;

MAIN:   CLI
        LDI R20,LOW(RAMEND)
        OUT SPL,R20
        LDI R20,HIGH(RAMEND)
        OUT SPH,R20
        LDI R20,0XFF
        OUT DDRA,R20
        OUT DDRC,R20
        LDI R20,0X20
        OUT GICR,R20
        LDI R20,0X40
        OUT MCUCSR,R20
        LDI R30,LOW(2*POLA) ; INIT POINTER Z
        LDI R31,HIGH(2*POLA)
        LDI R22,0               ; INIT COUNTER
        SEI
KK:         MOV R27,R22
        ANDI R27,0XF0           ; MENGAMBIL ENAMBELASAN
```

```
        LSR R27
        LSR R27
        LSR R27
        LSR R27
        MOV R30,R27
        LPM R2,Z
        OUT PORTA,R2
        MOV R27,R22
        ANDI R27,0X0F          ; MENGAMBIL SATUAN
        MOV R30,R27
        LPM R2,Z
        OUT PORTC,R2
        RCALL DELAY
        RJMP KK


DELAY:  CLI
        LDI R23,1
LP1:    LDI R24,4
LP2:    LDI R25,0
LP3:    DEC R25
        BRNE LP3
        DEC R24
        BRNE LP2
        DEC R23
        BRNE LP1
        SEI
        RET
        ;=================
INT_2:  CLI
        INC R22
        CPI R22,0X2D
        BRNE LK1
        LDI R22,0X2C
```

LK1: **SEI**

RETI

;==================

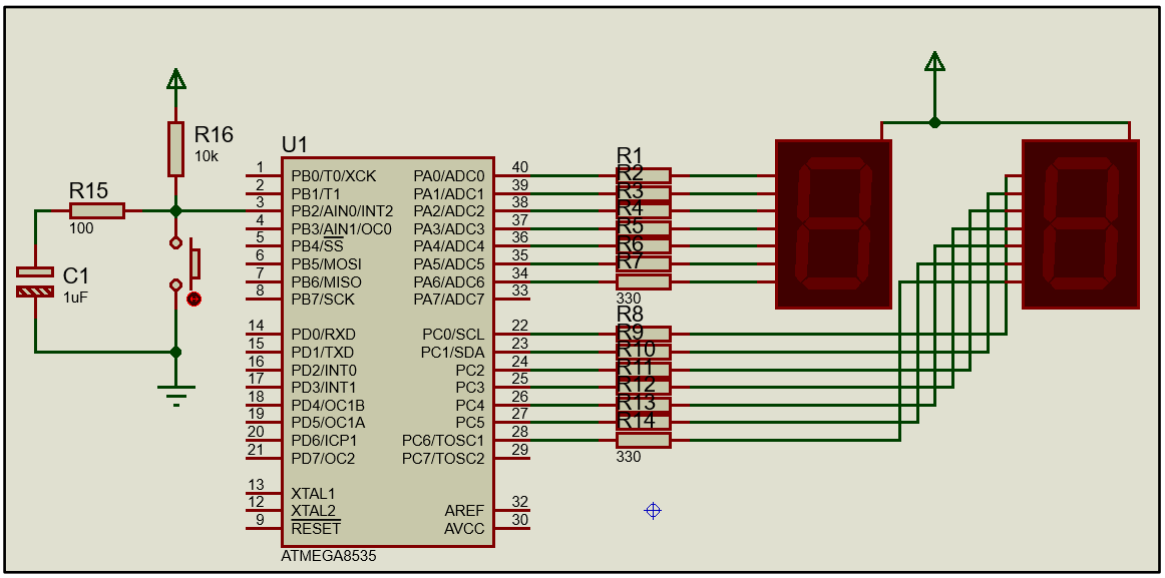
.ORG 0X80

POLA:

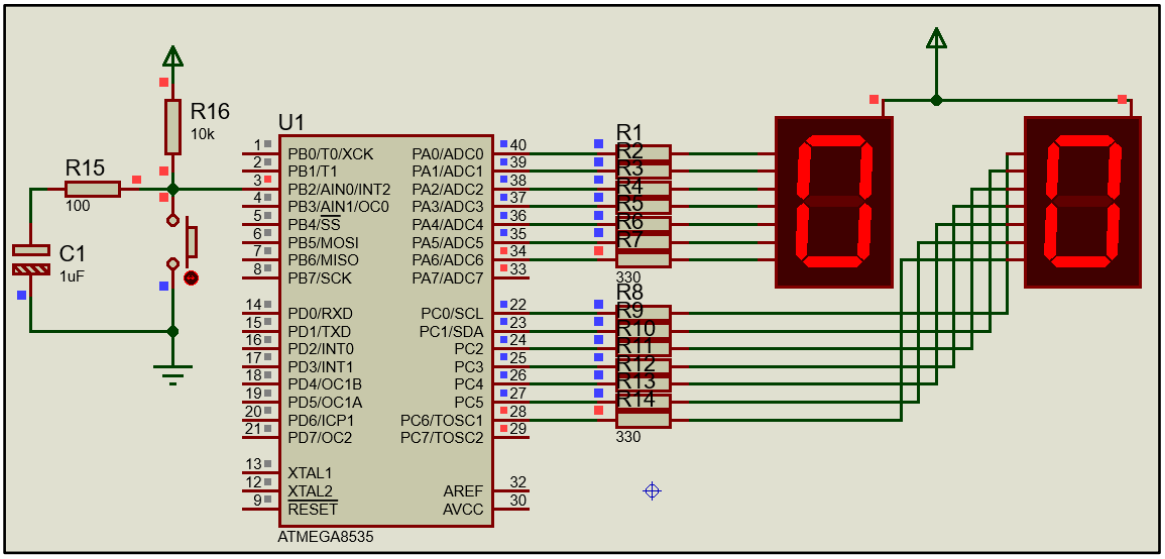
.DB 0XC0,0XF9,0XA4,0XB0,0X99,0X92,0X82,0XF8

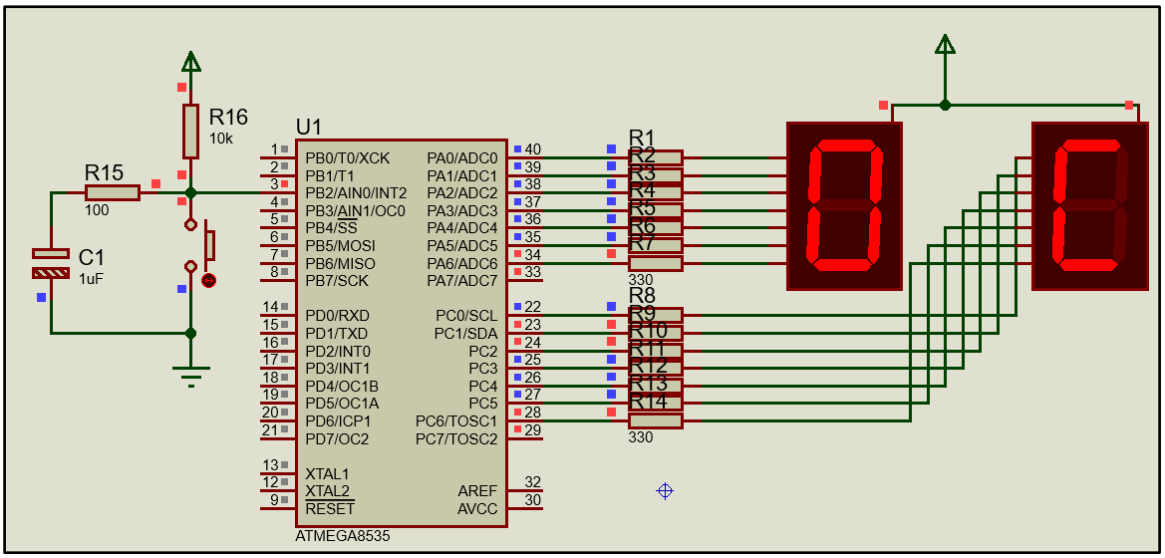
.DB 0X80,0X90,0X88,0X83,0XC6,0XA1,0X86,0X8E

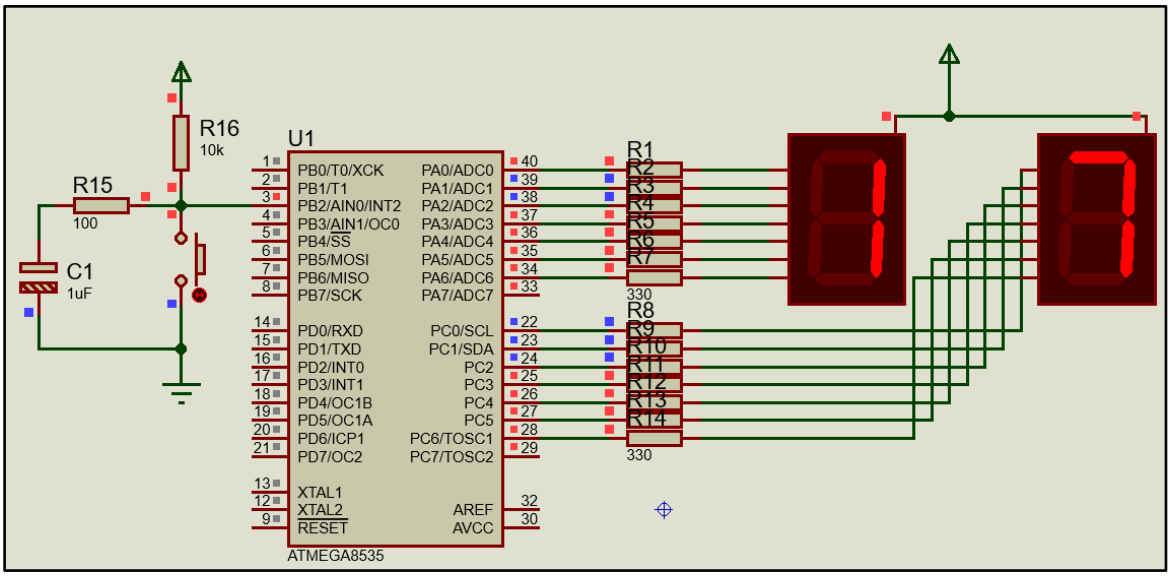
- **Rangkaian Simulasi**

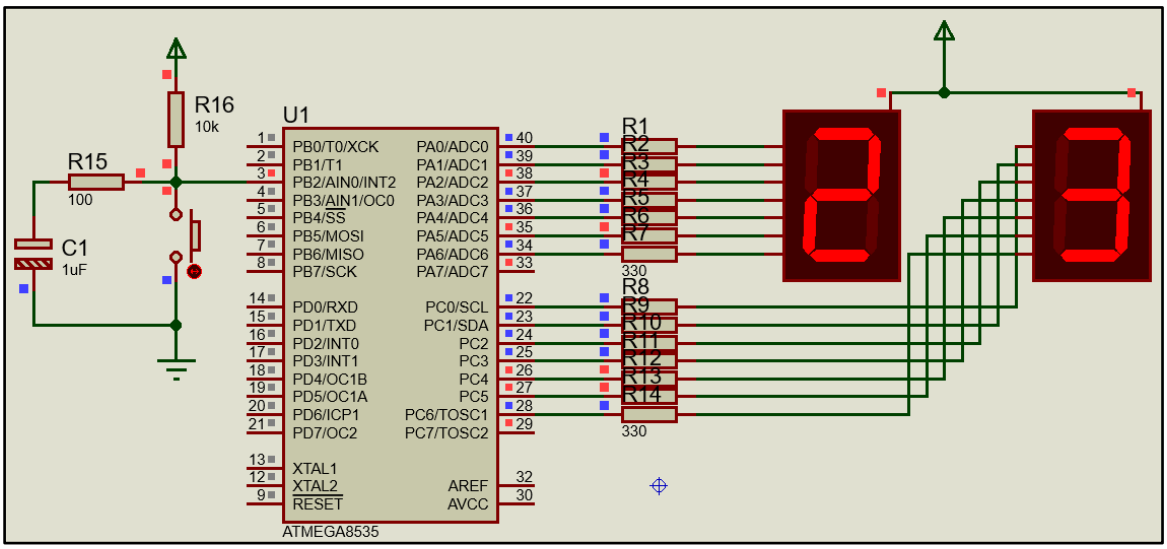


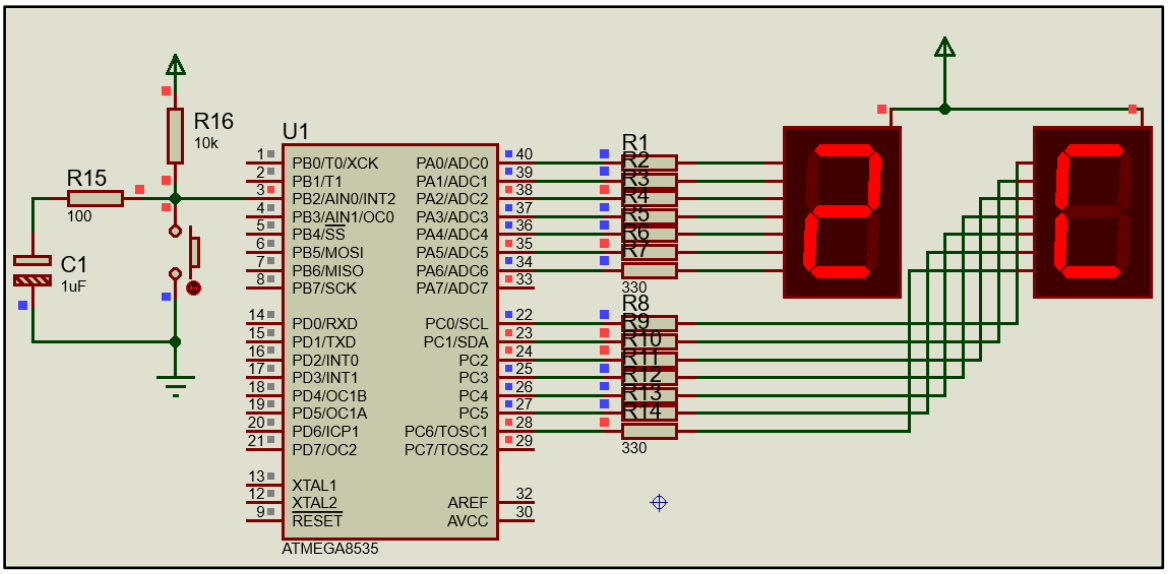
- **Ouput Program**




R16
10k
R15
100
C1
1uF
U1
PB0/T0/XCK
PB1/T1
PB2/AIN0/INT2
PB3/AIN1/OC0
PB4/SS
PB5/MOSI
PB6/MISO
PB7/SCK
PD0/RXD
PD1/TXD
PD2/INT0
PD3/INT1
PD4/OC1B
PD5/OC1A
PD6/ICP1
PD7/OC2
XTAL1
XTAL2
RESET
PA0/ADC0
PA1/ADC1
PA2/ADC2
PA3/ADC3
PA4/ADC4
PA5/ADC5
PA6/ADC6
PA7/ADC7
PC0/SCL
PC1/SDA
PC2
PC3
PC4
PC5
PC6/TOSC1
PC7/TOSC2
AREF
AVCC
ATMEGA8535
R1
R2
R3
R4
R5
R6
R7
330
R8
R9
R10
R11
R12
R13
R14
330


R16
10k
R15
100
C1
1uF
U1
PB0/T0/XCK
PB1/T1
PB2/AIN0/INT2
PB3/AIN1/OC0
PB4/SS
PB5/MOSI
PB6/MISO
PB7/SCK
PD0/RXD
PD1/TXD
PD2/INT0
PD3/INT1
PD4/OC1B
PD5/OC1A
PD6/ICP1
PD7/OC2
XTAL1
XTAL2
RESET
PA0/ADC0
PA1/ADC1
PA2/ADC2
PA3/ADC3
PA4/ADC4
PA5/ADC5
PA6/ADC6
PA7/ADC7
PC0/SCL
PC1/SDA
PC2
PC3
PC4
PC5
PC6/TOSC1
PC7/TOSC2
AREF
AVCC
ATMEGA8535
R1
R2
R3
R4
R5
R6
R7
330
R8
R9
R10
R11
R12
R13
R14
330


R16
10k
R15
100
C1
1uF
U1
PB0/T0/XCK
PB1/T1
PB2/AIN0/INT2
PB3/AIN1/OC0
PB4/SS
PB5/MOSI
PB6/MISO
PB7/SCK
PD0/RXD
PD1/TXD
PD2/INT0
PD3/INT1
PD4/OC1B
PD5/OC1A
PD6/ICP1
PD7/OC2
XTAL1
XTAL2
RESET
PA0/ADC0
PA1/ADC1
PA2/ADC2
PA3/ADC3
PA4/ADC4
PA5/ADC5
PA6/ADC6
PA7/ADC7
PC0/SCL
PC1/SDA
PC2
PC3
PC4
PC5
PC6/TOSC1
PC7/TOSC2
AREF
AVCC
ATMEGA8535
R1
R2
R3
R4
R5
R6
R7
330
R8
R9
R10
R11
R12
R13
R14
330



5. Seperti soal nomor 4, tapi untuk Down Counter.

- **Kode Program**

```
.include "m8535def.inc"
.org 0
RJMP MAIN

.ORG 0X012
RJMP INT_2
; Down-Counter Heksadesimal Modulo 45 (Menghitung secara berulang
; dari 2C sampai 0); INT0 berfungsi sebagai Counter Incrementer
; Menggunakan Seven Segment Common Anode,
; PROTEUS PROJECT: UP-CNT-INT0
;

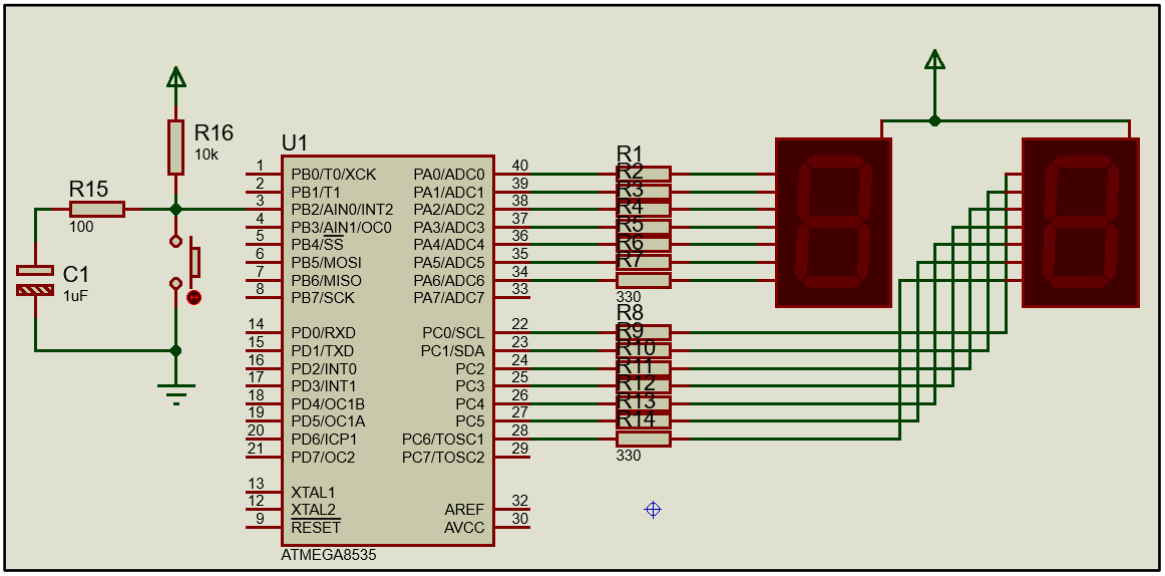
MAIN:   CLI
        LDI R20,LOW(RAMEND)
        OUT SPL,R20
        LDI R20,HIGH(RAMEND)
        OUT SPH,R20
        LDI R20,0XFF
        OUT DDRA,R20
        OUT DDRC,R20
```

```
        LDI R20,0X20
        OUT GICR,R20
        LDI R20,0X40
        OUT MCUCSR,R20
        LDI R30,LOW(2*POLA) ; INIT POINTER Z
        LDI R31,HIGH(2*POLA)
        LDI R22,0X2C            ; INIT COUNTER
        SEI
KK:         MOV R27,R22
        ANDI R27,0XF0           ; MENGAMBIL ENAMBELASAN
        LSR R27
        LSR R27
        LSR R27
        LSR R27
        MOV R30,R27
        LPM R2,Z
        OUT PORTA,R2
        MOV R27,R22
        ANDI R27,0X0F           ; MENGAMBIL SATUAN
        MOV R30,R27
        LPM R2,Z
        OUT PORTC,R2
        RCALL DELAY
        RJMP KK

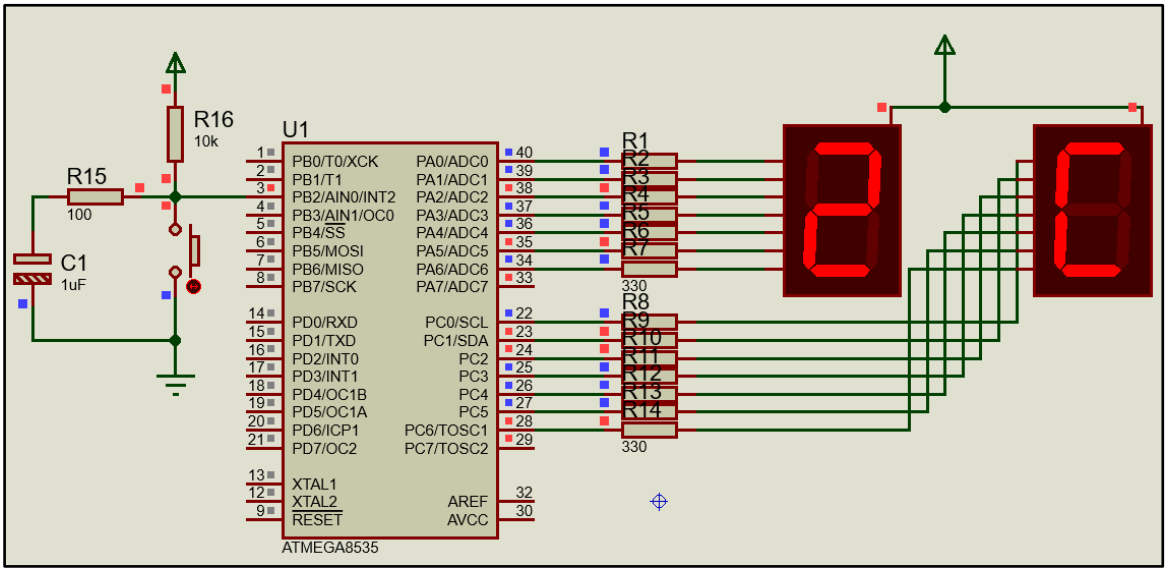

DELAY:  CLI
        LDI R23,1
LP1:    LDI R24,4
LP2:    LDI R25,0
LP3:    DEC R25
        BRNE LP3
        DEC R24
        BRNE LP2
```

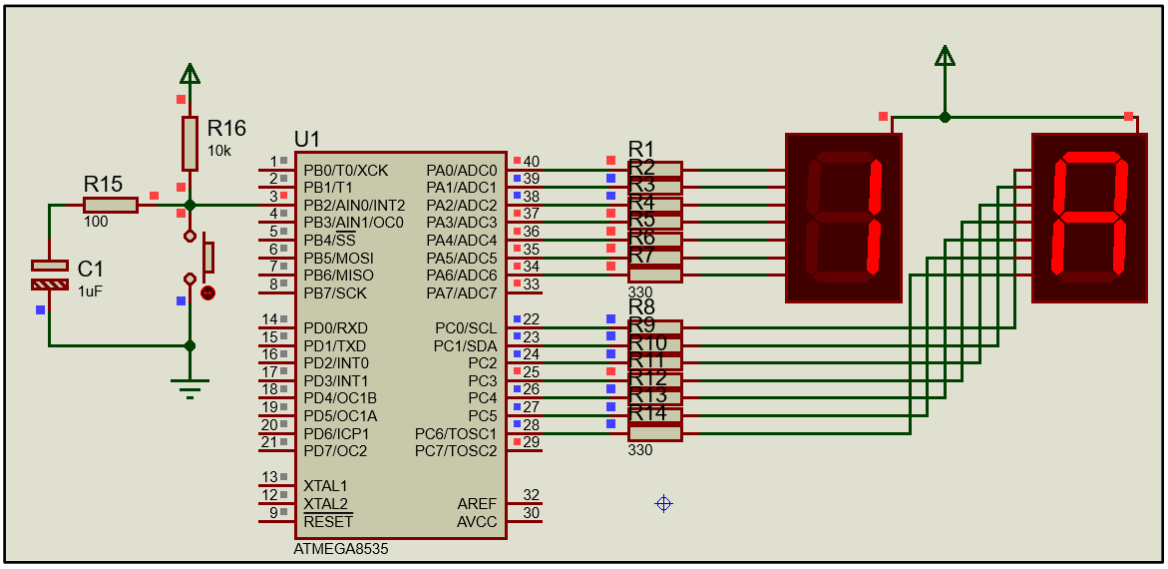
```
            DEC R23
            BRNE LP1
            SEI
            RET
            ;=================
INT_2:      CLI
            DEC R22
            CPI R22,0XFF
            BRNE LK1
            LDI R22,0X2C
LK1:        SEI
            RETI
;==================
.ORG 0X80
POLA:
.DB 0XC0,0XF9,0XA4,0XB0,0X99,0X92,0X82,0XF8
.DB 0X80,0X90,0X88,0X83,0XC6,0XA1,0X86,0X8E
```

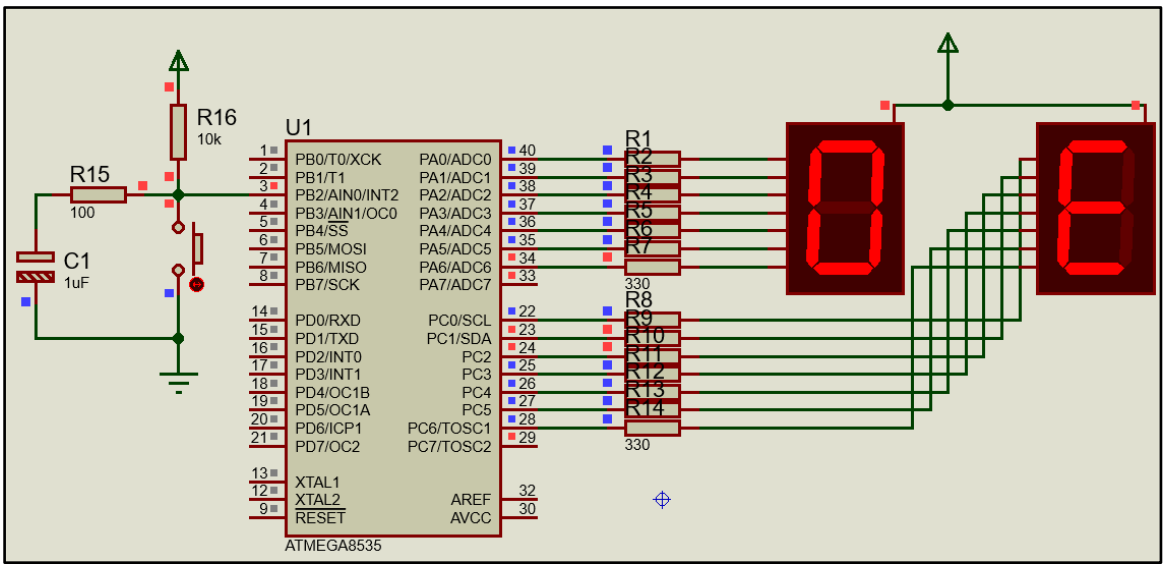
- **Rangkaian Simulasi**

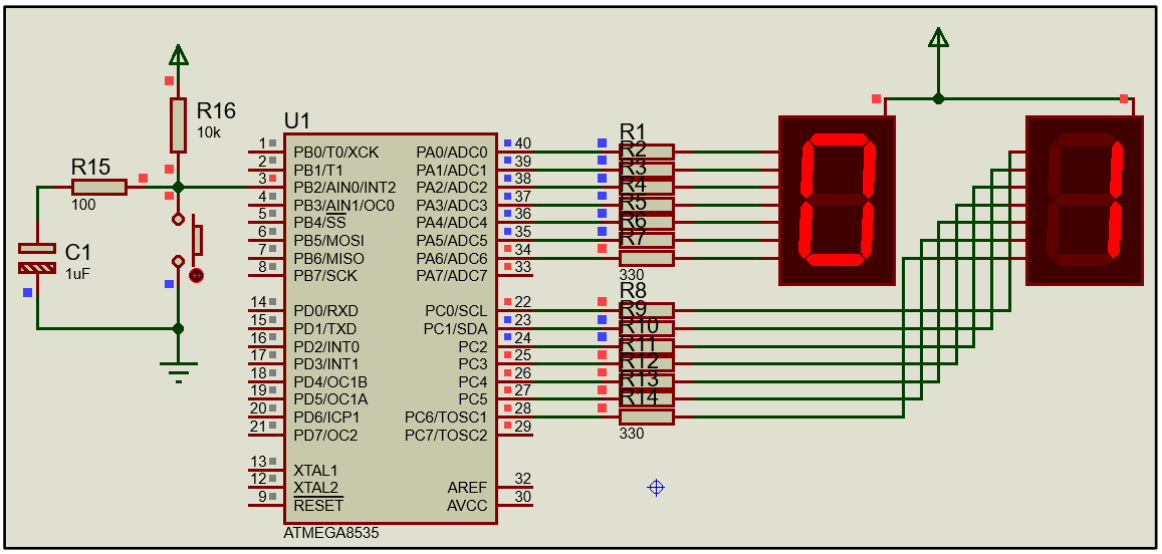


- **Ouput Program**









6. Rancang suatu Up/Down Counter Heksadesimal 2 digit dengan ketentuan :
   - INT0 digunakan untuk mengubah nilai counter
   - INT1 digunakan untuk menentukan nilai modulus counter, mod 21 (0 – 14) dan mod 35 (0 – 22)
   - INT2 digunakan untuk mengatur mode kerja counter

- **Kode Program**

```
.include "m8535def.inc"
.org 0
RJMP MAIN

.ORG 0X001
RJMP INT_0

.ORG 0X002
RJMP INT_1

.ORG 0X012
RJMP INT_2
; Up/Down-Counter Heksadesimal Modulo 21 dan 35
; INT0 berfungsi sebagai Counter Incrementer
INT1 mengubah modulo. INT2 mode kerja counter
```

```
; Menggunakan Seven Segment Common Anode.
; PROTEUS PROJECT: UP-CNT-INT0


MAIN:   CLI
        LDI R20,LOW(RAMEND)
        OUT SPL,R20
        LDI R20,HIGH(RAMEND)
        OUT SPH,R20
        LDI R20,0XFF
        OUT DDRA,R20
        OUT DDRC,R20
        LDI R20,0XE0
        OUT GICR,R20
        LDI R20,0X0F
        OUT MCUCR,R20
        LDI R20,0X40
        OUT MCUCSR,R20
        LDI R30,LOW(2*POLA) ; INIT POINTER Z
        LDI R31,HIGH(2*POLA)
        LDI R22,0               ; INIT COUNTER
        SEI
KK:         MOV R27,R22
        ANDI R27,0XF0           ; MENGAMBIL ENAMBELASAN
        LSR R27
        LSR R27
        LSR R27
        LSR R27
        MOV R30,R27
        LPM R2,Z
        OUT PORTA,R2
        MOV R27,R22
        ANDI R27,0X0F           ; MENGAMBIL SATUAN
        MOV R30,R27
```

```
        LPM R2,Z
        OUT PORTC,R2
        RCALL DELAY
        RJMP KK

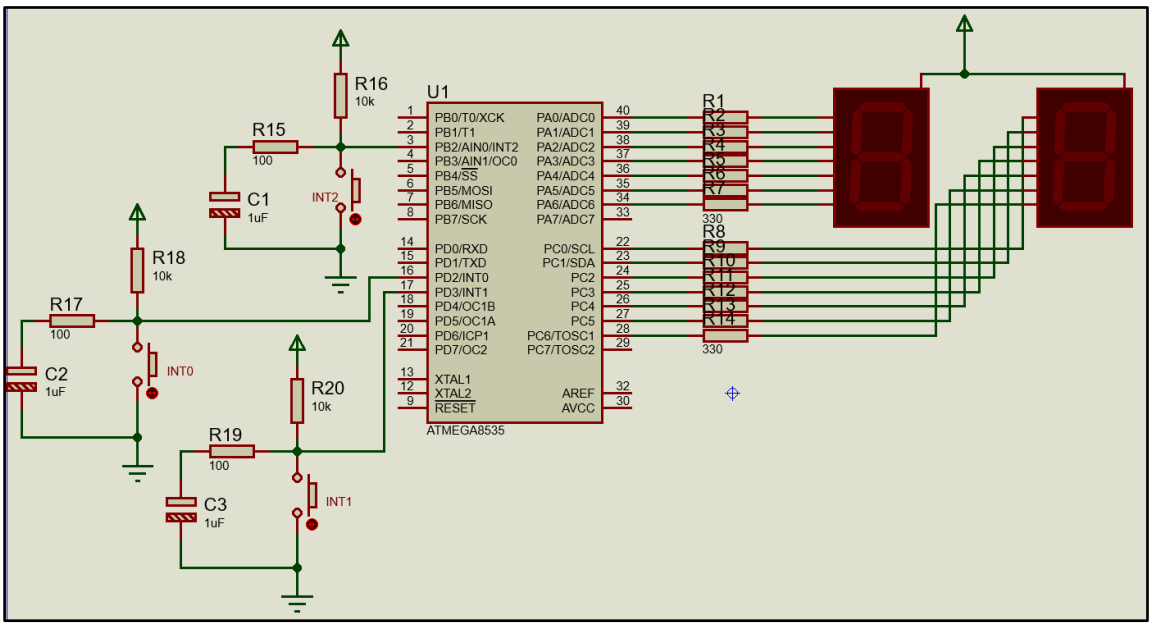

DELAY:  CLI
        LDI R23,1
LP1:    LDI R24,4
LP2:    LDI R25,0
LP3:    DEC R25
        BRNE LP3
        DEC R24
        BRNE LP2
        DEC R23
        BRNE LP1
        SEI
        RET
;=================
INT_0:  CLI
        MOV R21,R17
        ANDI R21,1
        BRNE TRN
        MOV R19,R18
        ANDI R19,1
        BRNE MOD35H
        INC R22
        CPI R22,0X15
        BRLT LK1
        LDI R22,0
        RJMP LK1
MOD35H: INC R22
        CPI R22,0X23
        BRNE LK1
```

```
            LDI R22,0
            RJMP LK1
TRN:        MOV R19,R18
            ANDI R19,1
            BRNE MD35H
            DEC R22
            CPI R22,0XFF
            BRNE LK1
            LDI R22,0X14
            RJMP LK1
MD35H:      DEC R22
            CPI R22,0XFF
            BRNE LK1
            LDI R22,0X22
LK1:        SEI
            RETI

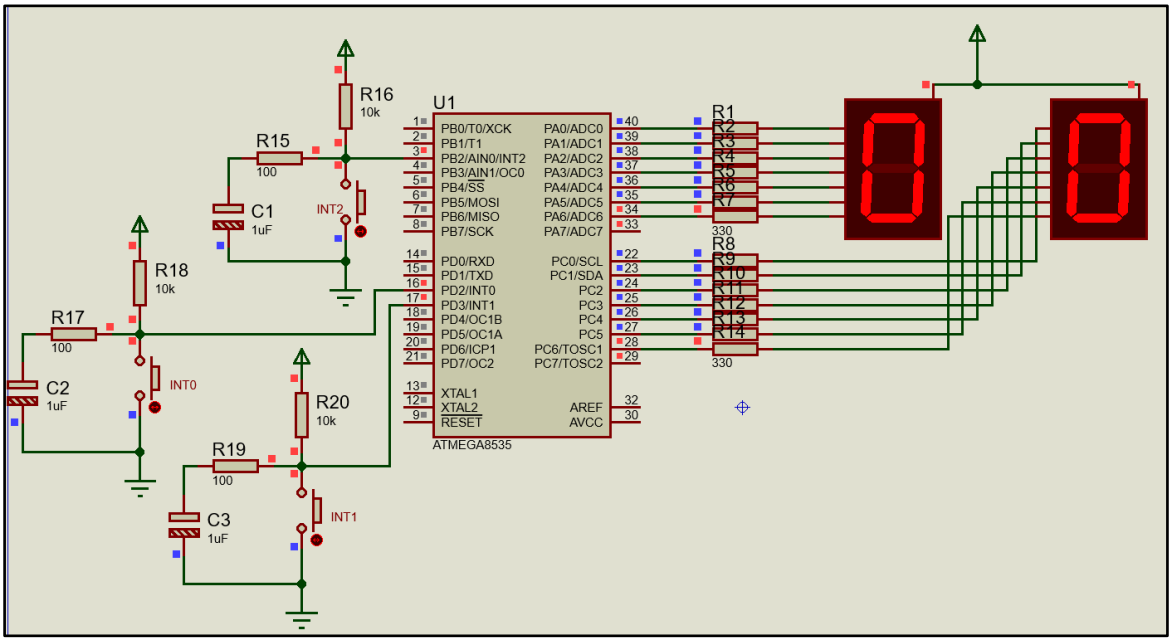
INT_1:      CLI
            INC R18
            SEI
            RETI

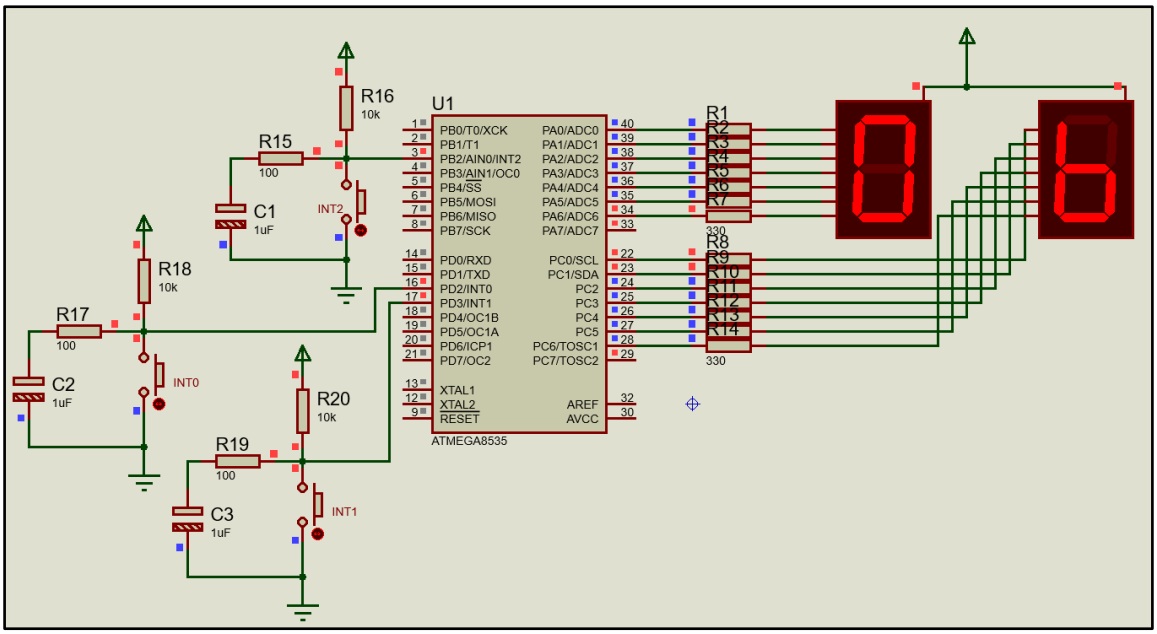
INT_2:      CLI
            INC R17
            SEI
            RETI
;==================
.ORG 0X80
POLA:
.DB 0XC0,0XF9,0XA4,0XB0,0X99,0X92,0X82,0XF8
.DB 0X80,0X90,0X88,0X83,0XC6,0XA1,0X86,0X8E
```

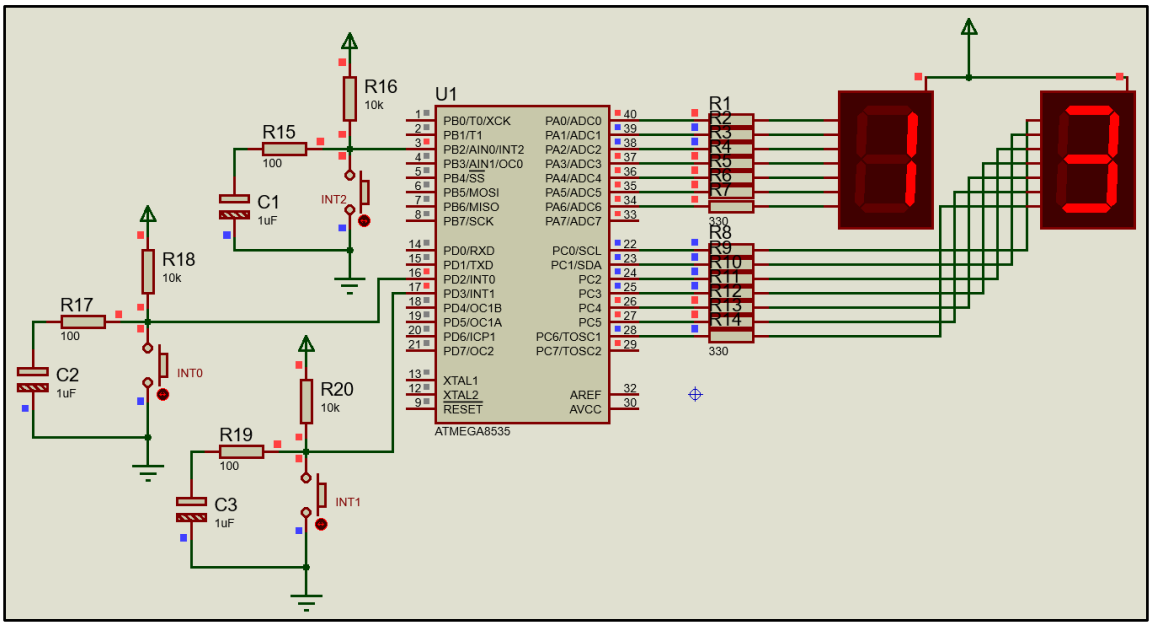
- **Rangkaian Simulasi**

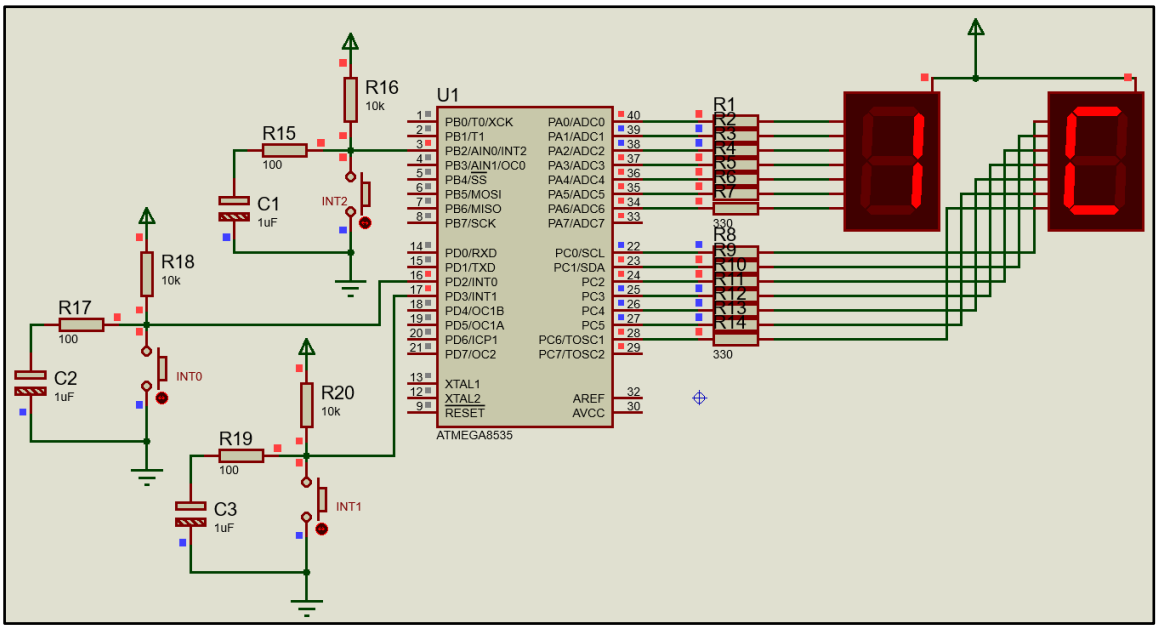


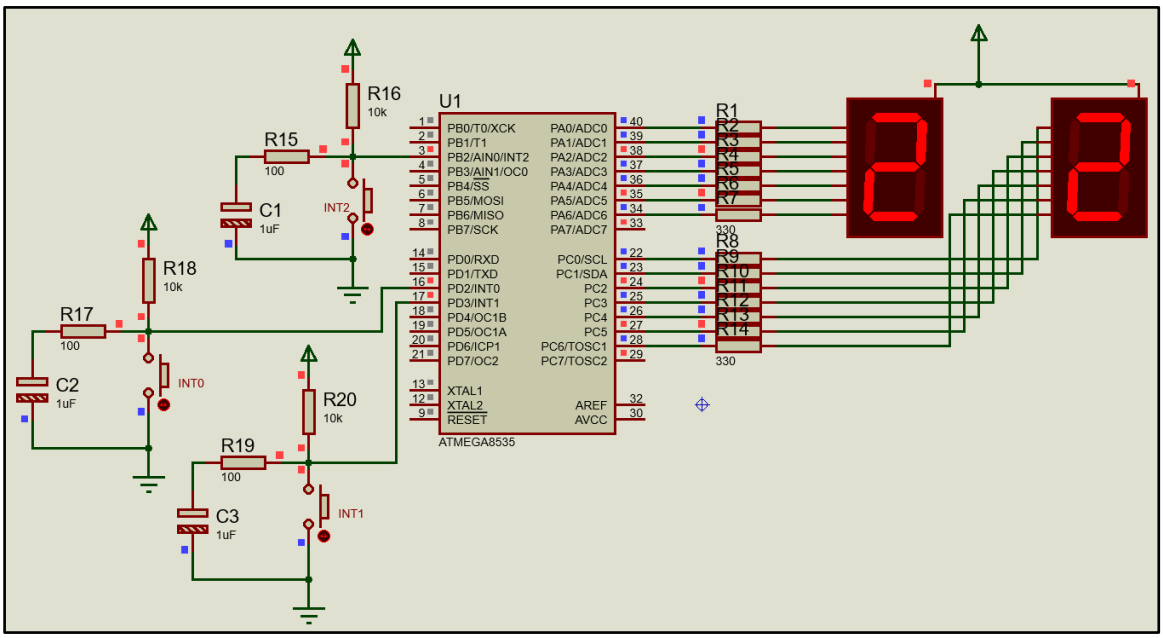
- **Output Program**




R16
10k
U1
R15
100
C1
1uF
INT2
R18
10k
R17
100
C2
1uF
INT0
R20
10k
R19
100
C3
1uF
INT1
PB0/T0/XCK
PB1/T1
PB2/AIN0/INT2
PB3/AIN1/OC0
PB4/SS
PB5/MOSI
PB6/MISO
PB7/SCK
PD0/RXD
PD1/TXD
PD2/INT0
PD3/INT1
PD4/OC1B
PD5/OC1A
PD6/ICP1
PD7/OC2
XTAL1
XTAL2
RESET
PA0/ADC0
PA1/ADC1
PA2/ADC2
PA3/ADC3
PA4/ADC4
PA5/ADC5
PA6/ADC6
PA7/ADC7
PC0/SCL
PC1/SDA
PC2
PC3
PC4
PC5
PC6/TOSC1
PC7/TOSC2
AREF
AVCC
ATMEGA8535
R1
R8
330
0
6


R16
10k
U1
R15
100
C1
1uF
INT2
R18
10k
R17
100
C2
1uF
INT0
R20
10k
R19
100
C3
1uF
INT1
PB0/T0/XCK
PB1/T1
PB2/AIN0/INT2
PB3/AIN1/OC0
PB4/SS
PB5/MOSI
PB6/MISO
PB7/SCK
PD0/RXD
PD1/TXD
PD2/INT0
PD3/INT1
PD4/OC1B
PD5/OC1A
PD6/ICP1
PD7/OC2
XTAL1
XTAL2
RESET
PA0/ADC0
PA1/ADC1
PA2/ADC2
PA3/ADC3
PA4/ADC4
PA5/ADC5
PA6/ADC6
PA7/ADC7
PC0/SCL
PC1/SDA
PC2
PC3
PC4
PC5
PC6/TOSC1
PC7/TOSC2
AREF
AVCC
ATMEGA8535
R1
R8
330
0
1
3


R16
10k
R15
100
C1
1uF
INT2
U1
PB0/T0/XCK
PB1/T1
PB2/AIN0/INT2
PB3/AIN1/OC0
PB4/SS
PB5/MOSI
PB6/MISO
PB7/SCK
PD0/RXD
PD1/TXD
PD2/INT0
PD3/INT1
PD4/OC1B
PD5/OC1A
PD6/ICP1
PD7/OC2
XTAL1
XTAL2
RESET
PA0/ADC0
PA1/ADC1
PA2/ADC2
PA3/ADC3
PA4/ADC4
PA5/ADC5
PA6/ADC6
PA7/ADC7
PC0/SCL
PC1/SDA
PC2
PC3
PC4
PC5
PC6/TOSC1
PC7/TOSC2
AREF
AVCC
ATMEGA8535
R1
R8
330
R18
10k
R17
100
C2
1uF
INT0
R20
10k
R19
100
C3
1uF
INT1


R16
10k
R15
100
C1
1uF
INT2
U1
PB0/T0/XCK
PB1/T1
PB2/AIN0/INT2
PB3/AIN1/OC0
PB4/SS
PB5/MOSI
PB6/MISO
PB7/SCK
PD0/RXD
PD1/TXD
PD2/INT0
PD3/INT1
PD4/OC1B
PD5/OC1A
PD6/ICP1
PD7/OC2
XTAL1
XTAL2
RESET
PA0/ADC0
PA1/ADC1
PA2/ADC2
PA3/ADC3
PA4/ADC4
PA5/ADC5
PA6/ADC6
PA7/ADC7
PC0/SCL
PC1/SDA
PC2
PC3
PC4
PC5
PC6/TOSC1
PC7/TOSC2
AREF
AVCC
ATMEGA8535
R1
R8
330
R18
10k
R17
100
C2
1uF
INT0
R20
10k
R19
100
C3
1uF
INT1